
Fast Reconstruction of Mixtures of Bernoulli Product Distributions

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Abstract

Mixtures of Bernoulli product distributions are a simple and widely used latent-variable model, with applications in e.g. recommendation systems, crowdsourcing, and medical data analysis. We consider the problem of reconstructing the mixture parameters from oracle access to its probability generating polynomial (PGP), for instance represented by a probabilistic generating circuit (PGC). We show that the parameters are uniquely identifiable for almost all mixtures, and give a randomized algorithm that exactly recovers the mixture weights and component marginals for mixtures of r Bernoulli product distributions over n variables using only $O(nr^2)$ oracle queries. The algorithm repeatedly applies restrictions to $O(r)$ variables, extracts low-degree coefficients, and then recovers the parameters using a moment-based tensor decomposition. To the best of our knowledge, this is the *first* exact reconstruction algorithm in this PGP oracle model with query complexity linear in n and polynomial in r .

1. Introduction

Mixtures of product distributions are a simple but surprisingly expressive latent-variable model for unsupervised learning. Each component is a product distribution, and mixing induces dependencies between coordinates through an unobserved label. This model has been used across a range of applications and also serves as a convenient testbed for developing and analyzing learning algorithms. In this work we focus on the computational question of how efficiently such mixtures can be reconstructed when the dimension n is large.

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Guiding Question. *Can mixtures of product distributions be reconstructed with only $O(n)$ oracle queries?*

More concretely, suppose an unknown distribution over $\{0, 1\}^n$ is a mixture of r product distributions. Is there an algorithm whose query complexity is essentially linear in n (assuming r is modest compared to n), while still efficiently recovering all mixture parameters?

Mixtures of product distributions are among the most classical and widely studied latent-variable models. Each mixture component captures coordinate-wise independence, while the mixture introduces correlations through an unobserved (hidden) variable. Beyond their foundational role in probability and statistics, such models arise naturally in many modern applications. For example, they appear in image recognition, population genetics, recommendation systems, medical imaging, and crowdsourcing, among others (Pritchard et al., 2000; Juan & Vidal, 2004; Tomozei & Massoulié, 2014; Dawid & Skene, 1979). In fact, it is known that *any* discrete distribution can be represented as a mixture of Bernoulli product components (Grim, 2006).

In this work, we revisit the learning problem from an algebraic/symbolic point of view. Instead of relying solely on sample access, we consider a *symbolic* setting in which the distribution is presented through its *probability generating polynomial*¹ (PGP) in a compact representation. This access allows us to compute low-degree multilinear coefficients efficiently (equivalently, low-order moments), and enables a reconstruction strategy that is explicitly coefficient-driven. This perspective connects distribution learning with polynomial reconstruction and suggests new routes to fast and explicit algorithms in structured access models. A potential application of such a scenario is the case when we have already learned a probabilistic generating circuit (PGC) (Zhang et al., 2021). While such a representation supports tractable marginalization, we cannot simply read off the parameters of the mixture from the PGC, since it might store the PGP in a different, more efficient way, for instance, using DPPs as subcircuits. Our work extends the tractable operations that are available for

¹Since the joint distribution has finite support, PGP is the same as the probability generating function (PGF).

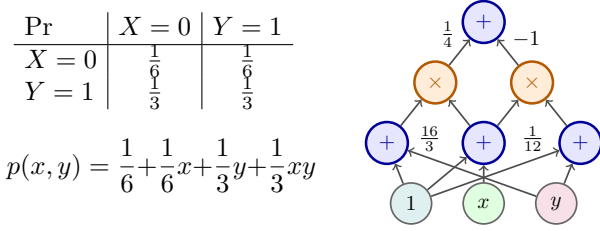


Figure 1. The tabular distribution represented as both: A probability generating polynomial p , and the probabilistic generating circuit.

PGC, in the spirit of Vergari et al. (2021), by adding parameter reconstruction with linear query complexity to the toolbox. This is essential as recovery of parameters helps us in answering all sorts of inference queries helping us reason about the data.

Model. We study mixtures of Bernoulli product distributions. An r -component mixture on variables $x_1, \dots, x_n$ can be written via its PGP as

$$f(x_1, \dots, x_n) = \sum_{i=1}^r \mu_i \prod_{j=1}^n ((1 - p_{ij})x_j + p_{ij}), \quad (1)$$

Here for all $i \in [r]$, $\mu_i \geq 0$ and $\sum_{i=1}^r \mu_i = 1$. Further, $p_{ij} \in [0, 1]$ for all $i \in [r]$, $j \in [n]$, and denote the marginal probability of $x_j = 1$ in component i . For algebraic convenience, we will often work with the equivalent re-parameterization $(1 - p_{ij})x_j + p_{ij} = 1 + t_{ij}x_j$. We can formalize the above PGP-access model via *probabilistic generating circuits (PGCs)*, which compute the polynomial f compactly. We are allowed to query the circuit beyond $\{0, 1\}^n$.

Why probabilistic generating circuits? A convenient way to represent a high-dimensional distribution over binary variables is through its *probability generating polynomial*, whose coefficients encode all joint probabilities, as shown in Figure 1. While this polynomial has exponentially many coefficients, recent work has shown that many structured distributions admit a compact representation as a *probabilistic generating circuit (PGC)*: a directed acyclic computation graph that evaluates the generating polynomial efficiently (Zhang et al., 2021; Harviainen et al., 2023; Agarwal & Bläser, 2024).

PGCs form a tractable probabilistic model class while also being expressive enough to subsume several classical tractable families including decomposable probabilistic circuits (PCs) and determinantal point processes (DPPs) (Zhang et al., 2021). PGCs sit within a growing view of tractable probabilistic models as succinct representations of multilinear polynomial semantics, where different polynomial encodings (network, likelihood, generating, etc.) can often be related by efficient transformations (Broadrick et al., 2024). Hence,

our results transfer to other means of access to the distribution probabilities, for instance, through a probabilistic circuit (PC) (Poon & Domingos, 2011; Choi et al., 2020). Furthermore, there has been substantial recent progress on improving both the learnability and scalability of such circuits, and practical algorithms for learning them from data are now available (Liu et al., 2024a; Gala et al., 2024; Zhang et al., 2025). A key advantage of these models is that they support exact inference queries efficiently, which makes them particularly attractive in applications where reliable probabilistic reasoning is important. Not only that, they have also been used in control and alignment of deep generative models (Liu et al., 2024b; Zhang et al., 2024) among other applications (Saad et al., 2021; Wedenig et al., 2024).

1.1. Our results

As our main result, we give an algorithm for recovering the parameters of a mixture of Bernoulli product distributions, as given by Equation 1. We assume that we have a query access to the PGP representing the distribution, for instance via PGCs. Our algorithm is designed for the “generic” setting. Gyllenberg et al. (1994) showed that mixtures of Bernoulli product distributions are nontrivially non-identifiable, i.e., there exists more than one set of parameter tuples (ignoring permutations of mixture components or coincident component distributions) that agree with the observed data. However, Allman et al. (2009) showed that the set of these non-identifiable parameters can be characterized through algebraic conditions and form a set of measure zero within the parameter space. (In algebraic geometry, this is called a Zariski-closed set, see Vakil, 2025.) Hence, we can “almost always” identify the parameters of models originating in practice.

Remark 1.1. In the generic setting, we assume that $p_{ij} \neq 0, 1$ for all $i \in [r]$ and $j \in [n]$ in Equation 1. This is a natural non-degeneracy assumption for mixtures of Bernoulli distributions. Formally, the set of parameters for which $p_{ij} \in \{0, 1\}$ for some i, j is given by a set of algebraic conditions, namely $(p_{ij} - 1)p_{ij} = 0$.

To the best of our knowledge, we provide the first algorithm for reconstruction of parameters of a mixture of Bernoulli product distributions such that the required number of queries is *linear* in the number of variables and *quadratic* in the number of mixtures r , assuming the aforementioned genericity conditions. In addition, the algorithm recovers the parameters of the distribution given by the PGP *exactly*. Secondly, while inference queries like marginalization are known to be tractable for PGCs (Zhang et al., 2021), we *expand the toolbox of efficient operations* on PGCs (equivalently, probabilistic

circuits), by demonstrating efficient parameter recovery in this setting. We now state our main result.

Theorem 1.2. *Given access to a PGP representing mixture of Bernoulli product distributions as shown in (1), we give an algorithm that reconstructs the exact parameters in $O(nr^2)$ queries in the generic setting.*

Remark 1.3. If we have oracle access to the PGP the algorithm has a runtime of $O\left(nr^{4+\delta}(\log r + \log \log \frac{1}{\epsilon}) + nr^{2+\omega}(\log r + \log \frac{1}{\epsilon})\right)$ for small enough $\delta, \epsilon > 0$. In general, the running time also includes an *additive* factor of $O(nsr^2)$ where s is the size of the representation of the PGP.

A key technical ingredient—of independent interest—is an algebraic method to recover missing entries from off-diagonal moments, avoiding matrix-completion style routines (see Proposition E.1).

Remark 1.4. We only assume query access to the PGP/PGC and can’t inspect the internal structure of the circuit. To our knowledge, no known algorithm using the circuit structure would improve the running time of our algorithm, since a PGC can store the PGP in a completely different way than the standard sum of products representation.

Remark 1.5. Determining the fan-in r from black-box access to f amounts to recovering r as a function of the hidden parameters (e.g., via the Jacobian with respect to μ_i and $t_{i,j}$). In the worst case, this task is computationally hard: even deciding whether f admits such a representation with fan-in at most r is known to be NP-hard (as it subsumes tensor-rank), as shown in Håstad (1990). These lower bounds rely on carefully constructed instances and do not reflect the generic regime we study. Indeed, as a corollary of our results, we can compute r in polynomial time (for $r \leq (n-1)/4$, in the generic setting) by enumerating candidate values of r , running our reconstruction algorithm, and certifying the output via polynomial identity testing.

Genericity. We prove reconstruction in a distribution-free *generic* sense: the relevant Jacobian (or, equivalently, an appropriate collection of its maximal minors) is not identically zero as a polynomial in the parameters, so the set of parameters where our identifiability condition fails lies in a proper algebraic variety. Equivalently, under any continuous parameter distribution, this non-degeneracy condition holds with probability 1. Importantly, this argument does not constrain the mixture weights μ_i .

1.2. Related works

Learning mixtures of product distributions. Learning such mixtures goes back to the classical EM algo-

rithm of Dempster et al. (1977), and early complexity-theoretic investigations by Kearns et al. (1994) and Dasgupta (1999). Efficient PAC-style algorithms for mixtures of discrete distributions were developed in Feldman et al. (2008), with subsequent improvements in more structured regimes, e.g., under separation assumptions (Chaudhuri & Rao, 2008), and via faster spectral/tensor methods (Chen & Moitra, 2019). Much of this literature focuses on recovering a distribution close to the truth rather than the exact parameters, and typically runs in time $n^{\text{poly}(r)}$, which quickly becomes impractical as r grows.

More recently, there has been progress on the *identification* problem of recovering a model close enough to the true model in the parameter space under additional assumptions. Gordon et al. (2021) gave a $2^{O(r^2)}n^{O(r)}$ algorithm for learning mixtures of r products from approximate multilinear moments under separability conditions, further refined in Gordon et al. (2024). In their work, they assume that the moments are given to them through an oracle access. In particular, under certain “separability conditions” they even give a $n2^{O(r^2)}$ running time. While this is linear in n , the algorithm gets impractical even for moderate r . In a different direction, Jain & Oh (2014) obtained a polynomial-time algorithm with running time $\text{poly}(n, r)$ under a structured setting where the components satisfy certain well-conditioning assumptions. While polynomial, the dependence on n is far from linear, and the approach relies on nontrivial linear-algebraic subroutines.

Algorithmically, a recurring theme is to exploit low-order moments (often second- and third-order) and perform a recovery step from these statistics, as in the general method-of-moments framework for latent-variable models (Anandkumar et al., 2014) using tensor decomposition tools such as the tensor power method (De Lathauwer et al., 2000). In the sample-based setting, moment estimation incurs sampling error; moreover, for Bernoulli product mixtures, Jain & Oh (2014) show that off-diagonal moment information alone does not directly provide the diagonal statistics needed for parameter recovery, motivating additional procedures such as matrix completion and least-squares based recovery.

A circuit reconstruction viewpoint. A second, complementary perspective comes from algebraic complexity theory. The probability generating polynomial of a mixture of Bernoulli product distributions admits a depth-3 sum-of-products representation (as in (1)), with top fan-in equal to the number of mixture components r . This links our setting to the literature on reconstructing structured polynomials from oracle (black-

box) access. In the *worst case*, however, reconstruction for general depth-3 circuits is known to be highly expensive even for constant r : for example, Bhargava et al. (2020, Theorem 1.6) gives a randomized algorithm with running time $n^{r^{r^{r^{10}}}}$ for depth-3 multilinear circuits. Subsequent results obtain polynomial-time guarantees in certain restricted regimes (Kayal & Saha, 2019), and quasipolynomial-time algorithms for constant r in greater generality (Saraf et al., 2025), but these bounds remain far from practical, even for moderately small fan-in r .

1.3. Our perspective and comparison

Our work sits at the intersection of two viewpoints—structured distribution learning and symbolic/algebraic reconstruction—but in a different access model: we assume the mixture is given through its *probability generating polynomial* in a compact representation, formalized as a PGP oracle or via PGCs (Zhang et al., 2021; Broadrick et al., 2024; Agarwal & Bläser, 2024) or similar tractable models. An application scenario is the case where we have already learned the PGC and now want to reconstruct the parameters, which one cannot read off from the circuit, since the learned circuits can have a structure that is different from the standard encoding as a sum of products. In particular, PGCs are even allowed to use negative weights, or can have determinantal circuits as subroutines (Zhang et al., 2021). Compared to worst-case circuit reconstruction, we benefit from working in a generic identifiability regime, which is a reasonable scenario for machine learning applications (Allman et al., 2009). Second, the PGP/PGC representation provides efficient symbolic access to low-degree coefficients. Once the PGC is learned, we have symbolic access to the PGP of the learned distribution. In this way, we work in an inference-oracle (noiseless) model: rather than receiving random samples, the learner can query the underlying generating object to obtain exact low-degree coefficient / moment information. This aligns more closely with classical query learning (Angluin, 1988) and with mixture-learning works that assume oracle access to the target distribution.

We utilize these low-degree multilinear coefficients (equivalently, lower-order moments) in two ways. First, unlike the standard sample-based setting, we can obtain the relevant moments *exactly* from the representation, eliminating sampling error in the reconstruction pipeline. Second, we show that (in a generic sense) projections to a small number of variables preserve the underlying mixture/circuit structure, allowing us to learn the model on low-dimensional restrictions and

stitch together the full solution across all n variables. In the same spirit as Jain & Oh (2014), we address the issue of recovering diagonal information from off-diagonal statistics. In the generic setting, this step admits a particularly efficient algebraic solution, avoiding the need for matrix completion or least-squares style procedures allowing us to prove Theorem 1.2.

Comparison with other linear time algorithms. (Rabani et al., 2014) study mixtures of *arbitrary (unstructured)* distributions over a discrete domain $[N]$, where N is the *domain size* (e.g., vocabulary size), and their positive results rely on *multi-snapshot* (“aperture”) samples (several i.i.d. draws from the *same* hidden component). In our setting, a single observation lies in $\{0, 1\}^n$, so the ambient domain has size 2^n ; thus, mapping our problem into their framework would correspond to $N = 2^n$, making their near-linear dependence on N exponential in n in our case.

Gordon et al. (2021; 2024) consider *source identification* for mixtures of r product distributions on n observable bits from conventional i.i.d. samples, taking as input empirical estimates of multilinear moments $g(S) = \mathbb{E}[X_S]$ up to order $|S| \leq 3r - 3$. Their reported (post-sampling) runtime is, in general, $2^{O(r^2)} \cdot n^{O(r)}$ arithmetic operations. Their strongest linear-in- n runtime regime requires a quantitative ζ -separation condition on the observables, and even then the runtime includes a factor $2^{O(r^2)}$, which renders the algorithm impractical even for moderate r .

2. Proof idea and algorithm overview

We are given oracle access to the probability generating polynomial (PGP) $f(x_1, \dots, x_n) = \sum_{i=1}^r \mu_i \prod_{j=1}^n ((1 - p_{ij})x_j + p_{ij})$, equivalently represented by a probabilistic generating circuit (PGC). Our goal is to exactly recover the weights $\{\mu_i\}_{i=1}^r$ and marginals $\{p_{ij}\}$ using as few oracle queries as possible. Note that if the parameters were arbitrary irrationals, then no practical algorithm could hope to recover them with infinite precision. Hence, we operate under the assumption that the parameters were specified up to some fixed precision ϵ .

Preprocessing and genericity. We first apply a preprocessing step to remove variables that do not appear in the PGP and to factor out any linear terms that are common to all mixture components, both of which can be detected using only $O(n)$ oracle queries; see Appendix A. Hereon, our guarantees hold under mild genericity assumptions (holding for almost all parameter choices) on the parameters p_{ij} , which rule out degenerate cases such as identical mixture components

or parameters lying on a proper algebraic subvariety. In particular, our algorithm doesn't require any restrictions on the mixture weights μ_i .

A key challenge now is that we cannot work with all n variables at once. Instead, our algorithm repeatedly restricts the PGP to only $m = O(r)$ variables (keeping the rest fixed), solves the reconstruction problem in this low-dimensional projection, and then stitches the solutions across blocks to recover all n coordinates. This approach requires resolving two conceptual issues:

(1) Small restrictions must preserve fan-in r . When we restrict to a subset of variables, it is not *a priori* clear that the resulting polynomial still corresponds to a mixture of r distinct product components; in principle, different components could collapse and the effective fan-in could drop below r . To rule this out, we study the coefficient map induced by a restriction and show that, for almost all parameter choices, the restricted PGP cannot be represented as a mixture of fewer than r products. Technically, we formalize this via a Jacobian rank separation argument: even though the Jacobian is *not full rank* in our setting, its rank is provably larger for an r -mixture than for an $(r - 1)$ -mixture, which certifies that the restriction preserves the mixture size generically, see Appendix C.

(2) Identifiability and exact recovery from low-order coefficients. Once a restriction preserves fan-in r , we must *recover* the parameters from that restricted instance. Oracle access to the PGP allows us to extract low-degree multilinear coefficients exactly via interpolation using only polynomially many queries in m (hence polynomial in r). These low-degree coefficients can be organized as second- and third-order moment objects of the restricted model, leading to a structured low-rank tensor of the form $T = \sum_{i=1}^r \mu_i u_i^{\otimes 3}$. We then apply a moment-based tensor decomposition routine (whitening followed by a tensor power method) to recover the vectors u_i and weights μ_i (up to permutation); see Section 4.2 and Algorithm 2. To justify correctness, we show in Appendix G that the resulting rank- r tensor decomposition is generically unique (up to permutation and scaling) via Kruskal's theorem, ensuring that the recovered components correspond to the true mixture parameters.

A technical obstacle in prior spectral/sample-based approaches is that the required low-rank moment matrix/tensor *cannot* be read off directly from observed moments because certain diagonal blocks are missing; these are typically handled via matrix completion or least-squares optimization. In contrast, in our generic setting we recover the missing diagonal statistics *algebraically* by exploiting the fact that the relevant mo-

ment matrix has rank at most r , so every $(r+1) \times (r+1)$ minor vanishes. Choosing a minor that contains exactly one unknown diagonal entry gives a single linear equation in that entry, which we can solve in closed form without any numerical optimization. See Section 4.1 and Algorithm 1.

Stitching across blocks. A single restriction to $m = O(r)$ variables only reveals the parameters on those coordinates, and even within one block the recovered components are *a priori* defined only up to permutation (and scaling). However, it can be shown that the relevant rank- r tensor decomposition is generically unique, so the restricted parameters are well-defined up to a permutation of components. You can find the proof in Appendix G. To recover the full matrix (p_{ij}) over all n variables, we partition the variables into blocks of size $m = O(r)$ and reconstruct the restricted parameters blockwise. Since each block recovery is only defined up to a permutation of components, we align these permutations globally using either the mixture weights (when distinct) or by adding a small number of anchor variables to later blocks to avoid collisions (see Appendix H for the full technical argument). This yields exact global parameter recovery with query complexity linear in n and per-block cost polynomial in r .

3. Small restrictions preserve mixture size

In this work, we provide an efficient randomized algorithm to learn the parameters of the distribution. One key component of the algorithm is to project down to a few variables and then learn the parameters from that small projection. For this strategy to work, we must ensure that such restrictions do not collapse the mixture structure—in particular, that the fan-in remains r . We show below that such a structural relation always holds in the generic setting.

3.1. Small restriction

After preprocessing the input (see Appendix A), we may assume w.l.o.g. that f has no nontrivial common factor and that every variable appears in f .

To simplify the parametrization, we apply the shift $x_j \mapsto x_j + 1$ for all $j \in [n]$ in (1). Since we have oracle access to the PGP, we can evaluate the shifted polynomial by querying f at translated inputs. Define $\tilde{f}(x_1, \dots, x_n) := f(x_1 + 1, \dots, x_n + 1) = \sum_{i=1}^r \mu_i \prod_{j=1}^n ((1 - p_{ij})x_j + 1)$. For convenience, we write $t_{ij} := 1 - p_{ij}$, so that each factor becomes $t_{ij}x_j + 1$. Note that after this shift the linear forms are no longer normalized (their coefficients need not sum to 1), but this normalization is not required for our arguments.

Next choose an *alive* set $A \subseteq [n]$ of size m , and let

$F := [n] \setminus A$ be the set of *fixed* variables. We **emphasize** that the choice of A can be completely deterministic; our guarantees rely on the genericity over the parameters $\{t_{ij}\}$ rather than on the randomness in the restriction. Permuting indices if needed, assume $A = \{1, 2, \dots, m\}$. We set the fixed variables to 1, i.e., $x_F = \mathbf{1} := (1)_{j \in F}$, and obtain an m -variate polynomial $h_A(x_1, \dots, x_m) := \tilde{f}(x_1, \dots, x_m, 1, \dots, 1) = \sum_{i=1}^r \mu_i \prod_{j=1}^m (t_{ij} x_j + 1)$. From now on we relabel h_A as h , so we work with

$$h(x_1, \dots, x_m) = \sum_{i=1}^r \mu_i \prod_{j=1}^m (t_{ij} x_j + 1), \quad (2)$$

where $0 < t_{ij} < 1$ (by Remark 1.1).

The polynomial h is multilinear in $x_1, \dots, x_m$. Once we have h , we compute symbolic rank of the *Jacobian* (see Appendix B for a primer) to argue about the preservation of mixture size.

Intuition. Fix an alive set A of size m and view the coefficient vector $C = (c_S)_{S \subseteq [m]}$ of h as a function of the hidden parameters $V := \{\mu_i\}_{i \in [r]} \cup \{t_{ij}\}_{i \in [r], j \in [m]}$. This defines a polynomial map from parameters to observable coefficients. An r -mixture has more degrees of freedom than an $(r-1)$ -mixture, so generically its image has larger local dimension. We certify this separation by lower-bounding the symbolic rank of the Jacobian: generically, an r -mixture produces a *strictly higher* Jacobian rank than any $(r-1)$ -mixture under the same restriction, so the restricted instance still has mixture size r .

For any $S \subseteq [m]$, the coefficient of the monomial $x_S := \prod_{j \in S} x_j$ is given by $c_S = \sum_{i=1}^r \mu_i \prod_{j \in S} t_{ij}$. Let $C = (c_S)_{S \subseteq [m]} \in \mathbb{K}^{2^m}$ denote the vector of all coefficients. We only concern ourselves with real distributions and hence $\mathbb{K} = \mathbb{R}$. For each $i \in [r]$ and $S \subseteq [m]$,

$$\frac{\partial c_S}{\partial \mu_i} = \prod_{j \in S} t_{ij}, \quad \frac{\partial c_S}{\partial t_{i\ell}} = \mu_i \cdot \mathbf{1}_{\{\ell \in S\}} \prod_{j \in S \setminus \{\ell\}} t_{ij}.$$

Viewing the coefficients $\{c_S\}$ as polynomials in the parameter set V , we form the Jacobian matrix J_r (as defined in Equation 7) with 2^m rows indexed by $S \subseteq [m]$ and $r(m+1)$ columns indexed by variables $v \in V$, with entries $J_{S,v} := \frac{\partial c_S}{\partial v}$.

Let M_r denote the submatrix of J_r obtained by restricting to the columns indexed by $\{t_{i\ell}\}_{i \in [r], \ell \in [m]}$ and to the rows indexed by $S \neq \emptyset$. Thus M_r has size $(2^m - 1) \times (mr)$. In J_r , the μ -columns contribute at most r additional rank, while the row $S = \emptyset$ contributes one to the rank (because $\partial c_\emptyset / \partial \mu_i = 1$ for all i and $\partial c_\emptyset / \partial t_{i\ell} = 0$). Therefore,

$$1 + \text{rank}(M_r) \leq \text{rank}(J_r) \leq r + \text{rank}(M_r). \quad (3)$$

Hence, it suffices to determine $\text{rank}(M_r)$. In fact, we can show that $\text{rank}(M_r) = rm$ in the generic setting (see Appendix C for the full proof). Thus, we get

$$rm + 1 \leq \text{rank}(J_r) \leq rm + r. \quad (4)$$

Separating r from $r-1$. Once we have the above relation, we can separate J_r from J_{r-1} using the rank bounds. For $m \geq r+1$, the lower bound (4) gives $\text{rank}(J_r) \geq rm + 1$. On the other hand, applying (3) to an $(r-1)$ -mixture yields $\text{rank}(J_{r-1}) \leq (r-1) + \text{rank}(M_{r-1}) \leq (r-1) + (r-1)m$, since M_{r-1} has at most $(r-1)m$ columns. Because $m \geq r+1$, we have $rm + 1 > (r-1)m + (r-1)$, and hence the Jacobian ranks of the r - and $(r-1)$ -mixture models are separated. This leads us to the following proposition:

Proposition 3.1. *For $m \geq r+1$, in the generic setting, restricting to any alive set of size m does not reduce the top fan-in of the PGC, i.e., the mixture size remains r .*

4. Moment methods to recover parameters

Anandkumar et al. (2014) introduced the method-of-moments framework for estimating parameters in a broad class of latent-variable models. A key observation is that, for many such models, low-order moments (or cross-moments) admit a *low-rank* tensor structure. For instance, in the single-topic model, these moments form a symmetric tensor and, in certain regimes, even a symmetric *orthogonally decomposable* tensor (cf. Anandkumar et al. (2014, Theorem 3.1)). Concretely, the second- and third-order moments take the form

$$M_2 = \sum_{i=1}^r \mu_i \alpha_i \otimes \alpha_i, \quad (5)$$

$$M_3 = \sum_{i=1}^r \mu_i \alpha_i \otimes \alpha_i \otimes \alpha_i, \quad (6)$$

where $\mu_i \in \mathbb{R}$, $\alpha_i \in \mathbb{R}^k$ for all $i \in [r]$, for some $k \in \mathbb{N}$.

To recover the parameters from these moments, one typically assumes a mild non-degeneracy condition: the vectors $\alpha_1, \dots, \alpha_r$ are linearly independent (and $\mu_i \neq 0$). This implies that $\text{rank}(M_2) = r$, and moreover that $\text{rank}_{\text{CP}}(M_3) = r$, where $\text{rank}_{\text{CP}}(\cdot)$ denotes CP rank (see Kolda & Bader (2009)). When such non-degeneracy conditions fail, learning is conjectured to be computationally hard; see Mossel & Roch (2005); Moitra & Valiant (2010).

Remark 4.1. The linear independence of $\alpha_1, \dots, \alpha_r$ holds *generically*: the set of parameter choices for which the vectors become linearly dependent is characterized by the vanishing of all $r \times r$ minors of the

$k \times r$ matrix $(\alpha_1 | \dots | \alpha_r)$, and hence forms a proper algebraic subset. In particular, this condition holds in our setting as well.

The recovery task can be reduced to a tensor decomposition problem. For M_2 , (5) corresponds to the eigendecomposition of a rank- r symmetric matrix. While matrix eigendecompositions are well understood, tensor decompositions are not unique in general. However, after *whitening* using M_2 , the third-order moment becomes an orthogonally decomposable symmetric tensor, and its components can be recovered using the tensor power method (and its robust variants) as in Anandkumar et al. (2014). In our work, we use this tensor-decomposition routine as a black box after extracting M_2 and M_3 exactly from the PGP/PGC representation. We now briefly outline the full procedure.

4.1. Extracting moments using PGC

Recall that for any random restriction of size m , we can view the restricted distribution as encoded by the restricted PGP h (cf. Equation 2). As discussed above, throughout we assume that the vectors $\alpha_1, \dots, \alpha_r$ are linearly independent.

In the polynomial h , the degree-two terms are exactly the coefficients of monomials $x_i x_j$, and the degree-three terms are exactly the coefficients of monomials $x_i x_j x_k$ with *pairwise distinct* indices. Given a PGP such coefficients can be recovered by a standard interpolation procedure (see Appendix D). Hence, using Lemma D.1, we can recover all such coefficients in time $O(m^3)$. Thus, we obtain all entries $(M_2)_{i,j}$ for $i \neq j$ and all entries $(M_3)_{a,b,c}$ for pairwise distinct $a, b, c \in [m]$.

Remark 4.2. We query the PGP *only* for interpolating the coefficients on these small restrictions.

However, to apply the method of moments, we require the full objects M_2 and M_3 , and not just the “off-diagonal” entries. For example, the $(1, 1)$ entry of M_2 equals: $(M_2)_{1,1} = \sum_{\ell=1}^r \mu_\ell \alpha_{\ell,1}^2$. We cannot recover this quantity from interpolation, since the restricted PGP h is multilinear and hence does not contain monomials such as x_1^2 (or, more generally, monomials with repeated indices).

In sample-based settings, such missing diagonal (or repeated-index) statistics are typically handled using numerical estimation or optimization procedures. In fact, this issue is well-studied for mixtures of Bernoulli product distributions. Jain & Oh (2014) formulate recovery of the diagonal entries of M_2 as a matrix completion problem, and give an alternating minimization algorithm that can exactly recover the diagonal when

Algorithm 1 Moment extraction

- 1: **Input:** PGP $f(x_1, \dots, x_n)$
 - 2: **Output:** Moment objects M_2, M_3
 - 3: Use Lemma D.1 to obtain all off-diagonal entries of M_2 and all entries of M_3 with pairwise distinct indices.
 - 4: Use Lemma E.2 to recover the diagonal entries of M_2 (via vanishing $(r+1) \times (r+1)$ minors).
 - 5: Use vanishing minors of the mode-flattenings of M_3 to recover all its remaining entries.
-

provided exact off-diagonal entries. For M_3 , instead of computing it exactly, they estimate a whitened version of M_3 via a least-squares method using only the off-diagonal entries. While their algorithm runs in time $\text{poly}(n, r)$, the dependence is not linear, and it also requires M_2 to be sufficiently well-conditioned.

In contrast, we show that in the “generic” setting, we can recover the missing entries of both M_2 and M_3 *exactly* and *efficiently* using a purely algebraic procedure. Intuitively, since we have only r mixture components, we have $\text{rank}(M_2) \leq r$, and similarly every mode-flattening of M_3 has rank at most r . Therefore, all $(r+1) \times (r+1)$ minors of M_2 (and of each flattening of M_3) must vanish. By carefully choosing such submatrices so that they contain exactly one unknown entry, each vanishing determinant yields a linear equation in that unknown entry, which we can solve in closed form. If all the parameters had precision upto ϵ , then the determinant of any such matrix will have a precision of at most $O(r^{\frac{r}{2}} 2^{r \log \frac{1}{\epsilon}})$ (Bareiss, 1972). Hence, we only need to compute it upto $O(r(\log r + \log \frac{1}{\epsilon}))$ bits to compute each of the diagonal entries. This allows us to recover all missing entries of both M_2 and M_3 in total time $\tilde{O}(r^{3+\omega})$, where ω is the matrix multiplication exponent and $\tilde{O}(\cdot)$ is the standard variant of $O(\cdot)$ which subsumes polylogarithmic factors. We defer the technical details to Appendix E, and we summarize the moment extraction procedure in Algorithm 1.

4.2. Reconstructing the parameters

We first fix a restriction of size m . Then, as described above, we recover M_2 and M_3 exactly. We now briefly describe how we can learn the parameters using the approach of Anandkumar et al. (2014).

Since M_2 is a real symmetric $m \times m$ matrix of rank r , it admits an eigendecomposition $M_2 = \sum_{i=1}^r d_i u_i u_i^\top = U D U^\top$, where $U \in \mathbb{R}^{m \times r}$ has orthonormal columns, and $D \in \mathbb{R}^{r \times r}$ is diagonal with the (non-zero) eigenvalues $d_1, \dots, d_r$. We use this eigendecomposition to define a whitening matrix $W \in \mathbb{R}^{m \times r}$ such that $W^\top M_2 W = I_r$. Concretely, define $W = U D^{-1/2}$. Clearly, $W^\top M_2 W = I_r$.

Moreover, using $M_2 = \sum_{i=1}^r \mu_i \alpha_i \alpha_i^\top$, we have $W^\top M_2 W = \sum_{i=1}^r \mu_i (W^\top \alpha_i)(W^\top \alpha_i)^\top = \sum_{i=1}^r (\sqrt{\mu_i} W^\top \alpha_i)(\sqrt{\mu_i} W^\top \alpha_i)^\top$. Now set $\hat{\alpha}_i = \sqrt{\mu_i} W^\top \alpha_i \in \mathbb{R}^r$. Then, $I_r = \sum_{i=1}^r \hat{\alpha}_i \hat{\alpha}_i^\top$.

Since there are exactly r vectors in $\mathbb{R}^r$ and the above sum has full rank, the vectors $\hat{\alpha}_1, \dots, \hat{\alpha}_r$ form an orthonormal basis. Worst-case eigen decomposition bounds of $O(m^3)$ for an $m \times m$ matrix are well known (Golub & Van Loan (2013)).

Recall that $M_3 = \sum_{i=1}^r \mu_i \alpha_i^{\otimes 3}$, where M_3 is an $m \times m \times m$ tensor. We now whiten M_3 using W to obtain a smaller $r \times r \times r$ tensor: $\hat{M}_3 := \sum_{i=1}^r \mu_i (W^\top \alpha_i)^{\otimes 3} = \sum_{i=1}^r \frac{1}{\sqrt{\mu_i}} \hat{\alpha}_i^{\otimes 3}$.

To compute $\hat{M}_3$ from M_3 , we need to multiply M_3 with W along all three modes, i.e., $(\hat{M}_3)_{abc} = \sum_{i=1}^m \sum_{j=1}^m \sum_{\ell=1}^m W_{i,a} W_{j,b} W_{\ell,c} (M_3)_{ij\ell}$. Naively, this computation would take $O(m^3 r^3)$ time. However, we can speed up the whitening using a careful ordering of multiplications (see Appendix F), obtaining $\hat{M}_3$ in time $O(rm^3 + r^2 m^2 + r^3 m)$.

Once we have $\hat{M}_3$, we apply the robust tensor power method of Anandkumar et al. (2014). For any target accuracy $\epsilon > 0$ and any fixed constant $\delta > 0$, their algorithm runs in time $O(r^{5+\delta}(\log r + \log \log(1/\epsilon)))$ to extract all r eigenvector–eigenvalue pairs, even with repeated eigenvalues.

Finally, using Anandkumar et al. (2014, Theorems 4.1 and 4.3), the robust eigenvectors of $\hat{M}_3$ are precisely the vectors $\hat{\alpha}_i$, with corresponding eigenvalues $\lambda_i = 1/\sqrt{\mu_i}$. Thus, we can recover $\mu_i = 1/\lambda_i^2$. Moreover, since $\hat{\alpha}_i = \sqrt{\mu_i} W^\top \alpha_i$, we can recover α_i via the Moore–Penrose pseudoinverse of $W^\top$: $\alpha_i = (W^\top)^\dagger (1/\sqrt{\mu_i} \hat{\alpha}_i)$.

Here we use that each α_i lies in the column span of M_2 , so the above inversion is well-defined on the relevant subspace. It is known that the complexity of computing the pseudoinverse of an $m \times r$ matrix (with $m \geq r$) is $O(mr^2)$ (Golub & Van Loan (2013)).

Combining all steps, the runtime for these linear-algebraic operations is $O(m^3 + (rm^3 + r^2 m^2 + r^3 m) + r^{5+\delta}(\log r + \log \log 1/\epsilon) + mr^2)$. Ignoring polylogarithmic factors and for sufficiently small $\delta, \epsilon > 0$, this is $\tilde{O}(rm^3 + r^2 m^2 + r^3 m + r^{5+\delta})$. Since $m = O(r)$ suffices, the total running time per restriction is $\tilde{O}(r^{5+\delta})$ upto log factors. We briefly describe the procedure in Algorithm 2.

4.3. Running time analysis

Using the algorithm of Anandkumar et al. (2014) and our preprocessing step (see Appendix A), we have

Algorithm 2 Recovering parameters from moments

- 1: **Input:** Moment objects M_2, M_3 .
- 2: **Output:** Parameters $\{(\mu_i, \alpha_i)\}_{i=1}^r$.
- 3: Compute the eigendecomposition $M_2 = UDU^\top$ and set $W = UD^{-1/2}$.
- 4: Whiten M_3 to obtain $\hat{M}_3 = M_3(W, W, W)$.
- 5: Run the tensor power method on $\hat{M}_3$ to obtain robust eigenpairs $(\lambda_i, \hat{\alpha}_i)$.
- 6: Set $\mu_i = 1/\lambda_i^2$ and recover $\alpha_i = (W^\top)^\dagger (\hat{\alpha}_i/\sqrt{\mu_i})$.

three main steps: first we recover M_2, M_3 fully after extracting the missing entries, we then obtain the whitening matrix W from M_2 and use it to produce $\hat{M}_3$, and we then finally run the tensor power method on $\hat{M}_3$ to recover the parameters μ_i, α_i . If we do not use projections, then in settings where n sufficiently dominates r , the first step takes $O(n^3)$ time, the second step takes $O(n^3 r)$ time, and the final step takes $O(r^{5+\delta}(\log r + \log \log 1/\epsilon))$. In total, we obtain a worst-case running time of $O(n^3 r)$, which is practically inefficient when n is large. On the other hand, Jain & Oh (2014) use a matrix completion approach to recover the diagonal entries of the lower-order moments and obtain a runtime that is polynomially dependent on n, r , and the condition number $\sigma_1(M_2)/\sigma_r(M_2)$, where $\sigma_i(M_2)$ denotes the i -th singular value of M_2 .

However, if we use our “genericity” conditions along with small projections, then using Proposition 3.1, Lemma E.2, and the subsequent discussion, we obtain a significantly faster algorithm. As we saw above, we only need $m = O(r)$. Analysing the runtime of all the steps in the oracle access model: $O(n)$ for preprocessing, $\tilde{O}(r^{2+\omega})$ for recovering the moments exactly for each restriction, and then $\tilde{O}(r^{5+\delta})$ for recovering μ_i, α_i for any restriction. Instead of running the above analysis on n -dimensional moments, we run it on restrictions of size m with $m = O(r)$, and hence we need to perform this procedure for only n/m blocks. Combining all this, we obtain a total running time of $\tilde{O}(n/m \cdot (r^{5+\delta} + r^{3+\omega}) + n) = \tilde{O}(n(r^{4+\delta} + r^{2+\omega}))$, for sufficiently small $\delta, \epsilon > 0$. In the general setting, where each query takes time $O(s)$ (where s is the size of the representation), the runtime becomes $\tilde{O}(nsr^2 + n(r^{4+\delta} + r^{2+\omega}))$.

4.4. Numerical stability analysis

In this work, we recover the parameters of the distribution exactly, assuming they were specified up to some fixed precision ϵ . Indeed, if the parameters were arbitrary irrationals, then no practical algorithm could hope to recover them with infinite precision. The main reconstruction step in our algorithm uses the tensor

Algorithm 3 Main Algorithm

```

1: Input: PGP of  $f$ 
2: Output: Parameters  $\mu_i, \alpha_{ij}$  for all  $i \in [r], j \in [n]$ .
3: Use Algorithm 4 to preprocess the input.
4: Set  $m = 4r + 1$ .
5: for  $i$  from 1 to  $\lfloor \frac{n}{m} \rfloor$  do
6:   Pick restriction  $S_i$  of size  $m$  containing variables
      $x_i, \dots, x_{i+m-1}$  such that  $r - 1$  "identifying" variables
     (using Lemma H.2) are always selected except when
      $i = 1$ . {For  $i = \lfloor \frac{n}{m} \rfloor$  pick all the variables until  $x_n$ .}
7:   Use Algorithm 1 to recover the moments.
8:   Use Algorithm 2 to recover the parameters  $\mu_i, \alpha_{ij}$  for
      $i \in [r], j \in S$ .
9: end for
10: Stitch together the parameters recovered in each  $S_i$  to
    get complete reconstruction.

```

power method as used in Anandkumar et al. (2014). In their work, they show that the method has a quadratic rate of convergence if we start with a "good" vector. Formally, if e_t denotes the error at iteration t , then one has a recurrence of the form $\|e_{t+1}\| \leq C\|e_t\|^2$ for some constant $0 < C < 1$. In particular, the error decays doubly-exponentially with t . Hence, we get the $\log \log \frac{1}{\epsilon}$ factor in the runtime. However, we still need to get this "good" starting vector θ . Anandkumar et al. (2014) show that with probability $\geq 1 - \eta$, we can get such a θ in $r^{1+\delta} \log 1/\eta$ iterations. They also show that in $\log r$ iterations, starting from θ we get θ' which is a vector that is in the zone of quadratic convergence. Combining this with the $O(r^3)$ running time for each iteration for each of the r eigenvectors, we get a randomized algorithm with total running time of $O(r^{5+\delta}(\log r + \log \log \frac{1}{\epsilon}))$ which recovers all parameters exactly if they were specified up to ϵ precision.

Finally, we will also like to make the following remark. Gordon et al. (2021) show that source identification is possible if and only if at least $2r - 1$ random variables (RVs) have mixture probabilities that are mutually distinct and separated by ϵ . In this case, they give a $n^{O(r)}2^{r^2}$ query complexity algorithm for identification. Further, if they have all RVs ϵ -separated then they can give a $O(n2^{r^2})$ query complexity algorithm. In the same context, if we have all random variables as ϵ -separated our algorithm will return the parameters up to ϵ precision in $O(nr^2)$ query complexity. When instead only $2r - 1$ random variables are ϵ -separated, our method performs substantially better than Gordon et al. (2021). An implication of Lemma H.2 is that if one fixed variable, say X_1 , is known to be ϵ -separated, then keeping X_1 alive in every restriction already suffices to distinguish all mixture components. The main challenge is that we do not know beforehand which variables are the ϵ -separated ones. In Gordon et al.

(2021), this necessitates a brute-force search over all candidate subsets of size $2r - 1$, giving rise to the $n^{O(r)}$ overhead. By contrast, our algorithm needs only *one* good variable! So, it suffices to brute-force over the n possible choices of that variable. As a result, when only $2r - 1$ variables are ϵ -separated, we obtain query complexity $O(n^2r^2)$ where each parameter is recovered up to ϵ precision, significantly improving over their work.

5. Conclusion and future directions

In this work, to the best of our knowledge, we give the first algorithm that requires *linear* in n and *quadratic* in r queries to exactly learn the parameters defining a mixture of products of Bernoulli distributions.

We believe that there are many interesting future directions to consider. One of them being extending this setup to the noisy setting. Concretely, we can assume some bounded noise in the PGC outputs and then see how efficiently and closely we can reconstruct the parameters. Secondly, we want to explore reconstruction when, on a certain fraction of inputs, the PGP gives adversarial oracle outputs while being correct on the remaining queries. Finally, reconstruction for different types of distributions like mixtures of bounded-treewidth models, and sparse distributions represented by PGPs is also an interesting open question.

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Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

References

- Agarwal, S. and Bläser, M. Probabilistic generating circuits - demystified. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 329–342. PMLR, 21–27 Jul 2024.
- Allman, E. S., Matias, C., and Rhodes, J. A. Identifiability of parameters in latent structure models with

- many observed variables. *The Annals of Statistics*, 37 (6A):3099–3132, 2009.
- Anandkumar, A., Ge, R., Hsu, D. J., Kakade, S. M., Telgarsky, M., et al. Tensor decompositions for learning latent variable models. *J. Mach. Learn. Res.*, 15(1): 2773–2832, 2014.
- Angluin, D. Queries and concept learning. *Machine learning*, 2(4):319–342, 1988.
- Bareiss, E. H. Computational solutions of matrix problems over an integral domain. *IMA Journal of Applied Mathematics*, 10(1):68–104, 1972.
- Bhargava, V., Saraf, S., and Volkovich, I. Deterministic factorization of sparse polynomials with bounded individual degree. *Journal of the ACM (JACM)*, 67 (2):1–28, 2020.
- Broadrick, O., Zhang, H., and Van den Broeck, G. Polynomial semantics of tractable probabilistic circuits. In *Proceedings of the Fortieth Conference on Uncertainty in Artificial Intelligence*, volume 244 of *Proceedings of Machine Learning Research*, pp. 418–429. PMLR, 15–19 Jul 2024.
- Chaudhuri, K. and Rao, S. Learning mixtures of product distributions using correlations and independence. In *COLT*, volume 4, pp. 9–20, 2008.
- Chen, S. and Moitra, A. Beyond the low-degree algorithm: mixtures of subcubes and their applications. In *Proceedings of the 51st Annual ACM SIGACT Symposium on Theory of Computing*, pp. 869–880, 2019.
- Choi, Y., Vergari, A., and Van den Broeck, G. Probabilistic circuits: A unifying framework for tractable probabilistic models. *UCLA*. URL: <http://starai.cs.ucla.edu/papers/ProbCirc20.pdf>, pp. 6, 2020.
- Christandl, M. and Zuiddam, J. Tensor surgery and tensor rank. *Computational Complexity*, 28(1):27–56, 2019. doi: 10.1007/s00037-018-0164-8.
- Dasgupta, S. Learning mixtures of gaussians. In *40th annual symposium on foundations of computer science (Cat. No. 99CB37039)*, pp. 634–644. IEEE, 1999.
- Dawid, A. P. and Skene, A. M. Maximum likelihood estimation of observer error-rates using the em algorithm. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 28(1):20–28, 1979.
- De Lathauwer, L., De Moor, B., and Vandewalle, J. On the best rank-1 and rank-($r_1, r_2, \dots, r_n$) approximation of higher-order tensors. *SIAM journal on Matrix Analysis and Applications*, 21(4):1324–1342, 2000.
- Dempster, A. P., Laird, N. M., and Rubin, D. B. Maximum likelihood from incomplete data via the em algorithm. *Journal of the royal statistical society: series B (methodological)*, 39(1):1–22, 1977.
- Feldman, J., O’Donnell, R., and Servedio, R. A. Learning mixtures of product distributions over discrete domains. *SIAM Journal on Computing*, 37(5):1536–1564, 2008.
- Gala, G., de Campos, C., Vergari, A., and Quaeghebeur, E. Scaling continuous latent variable models as probabilistic integral circuits. *Advances in Neural Information Processing Systems*, 37:10959–10987, 2024.
- Golub, G. H. and Van Loan, C. F. *Matrix computations*. JHU press, 2013.
- Gordon, S., Mazaheri, B. H., Rabani, Y., and Schulman, L. Source identification for mixtures of product distributions. In *Conference on Learning Theory*, pp. 2193–2216. PMLR, 2021.
- Gordon, S. L., Jahn, E., Mazaheri, B., Rabani, Y., and Schulman, L. J. Identification of mixtures of discrete product distributions in near-optimal sample and time complexity. In *The Thirty Seventh Annual Conference on Learning Theory*, pp. 2071–2091. PMLR, 2024.
- Grim, J. Em cluster analysis for categorical data. In *Joint IAPR International Workshops on Statistical Techniques in Pattern Recognition (SPR) and Structural and Syntactic Pattern Recognition (SSPR)*, pp. 640–648. Springer, 2006.
- Gyllenberg, M., Koski, T., Reilink, E., and Verlaan, M. Non-uniqueness in probabilistic numerical identification of bacteria. *Journal of Applied Probability*, 31 (2):542–548, 1994.
- Harviainen, J., Peruvemba Ramaswamy, V., and Koivisto, M. On inference and learning with probabilistic generating circuits. In *Proceedings of the Thirty-Ninth Conference on Uncertainty in Artificial Intelligence*, volume 216 of *Proceedings of Machine Learning Research*, pp. 829–838. PMLR, 31 Jul–04 Aug 2023.
- Håstad, J. Tensor rank is NP-complete. *Journal of Algorithms*, 11(4):644–654, 1990.
- Jain, P. and Oh, S. Learning mixtures of discrete product distributions using spectral decompositions. In *Conference on Learning Theory*, pp. 824–856. PMLR, 2014.

- Juan, A. and Vidal, E. Bernoulli mixture models for binary images. In *Proceedings of the 17th International Conference on Pattern Recognition, 2004. ICPR 2004.*, volume 3, pp. 367–370. IEEE, 2004.
- Kayal, N. and Saha, C. Reconstruction of non-degenerate homogeneous depth three circuits. In *Proceedings of the 51st Annual ACM SIGACT Symposium on Theory of Computing*, pp. 413–424, 2019.
- Kearns, M., Mansour, Y., Ron, D., Rubinfeld, R., Schapire, R. E., and Sellie, L. On the learnability of discrete distributions. In *Proceedings of the twenty-sixth annual ACM symposium on Theory of computing*, pp. 273–282, 1994.
- Kolda, T. G. and Bader, B. W. Tensor decompositions and applications. *SIAM Review*, 51(3):455–500, 2009. doi: 10.1137/0707011X.
- Kruskal, J. B. Three-way arrays: rank and uniqueness of trilinear decompositions, with application to arithmetic complexity and statistics. *Linear algebra and its applications*, 18(2):95–138, 1977.
- Lee, J. M. *Introduction to Smooth Manifolds*, volume 218 of *Graduate Texts in Mathematics*. Springer, 2012.
- Leung, D., Drton, M., and Hara, H. Identifiability of directed Gaussian graphical models with one latent source. *Electronic Journal of Statistics*, 10(1):394 – 422, 2016. doi: 10.1214/16-EJS1111. URL <https://doi.org/10.1214/16-EJS1111>.
- Liu, A., Ahmed, K., and Van Den Broeck, G. Scaling tractable probabilistic circuits: A systems perspective. In Salakhutdinov, R., Kolter, Z., Heller, K., Weller, A., Oliver, N., Scarlett, J., and Berkenkamp, F. (eds.), *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 30630–30646. PMLR, 21–27 Jul 2024a. URL <https://proceedings.mlr.press/v235/liu24a.html>.
- Liu, A., Niepert, M., and den Broeck, G. V. Image inpainting via tractable steering of diffusion models. In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net, 2024b. URL <https://openreview.net/forum?id=NSIVHTbZBR>.
- Maitra, A. and Valiant, G. Settling the polynomial learnability of mixtures of gaussians. In *2010 IEEE 51st Annual Symposium on Foundations of Computer Science*, pp. 93–102. IEEE, 2010.
- Mossel, E. and Roch, S. Learning nonsingular phylogenies and hidden markov models. In *Proceedings of the thirty-seventh annual ACM symposium on Theory of computing*, pp. 366–375, 2005.
- Poon, H. and Domingos, P. Sum-product networks: A new deep architecture. In *2011 IEEE International Conference on Computer Vision Workshops (ICCV Workshops)*, pp. 689–690. IEEE, 2011.
- Pritchard, J. K., Stephens, M., and Donnelly, P. Inference of population structure using multilocus genotype data. *Genetics*, 155(2):945–959, 2000.
- Rabani, Y., Schulman, L. J., and Swamy, C. Learning mixtures of arbitrary distributions over large discrete domains. In *Proceedings of the 5th conference on Innovations in theoretical computer science*, pp. 207–224, 2014.
- Saad, F. A., Rinard, M. C., and Mansinghka, V. K. Sppl: probabilistic programming with fast exact symbolic inference. In *Proceedings of the 42nd acm sigplan international conference on programming language design and implementation*, pp. 804–819, 2021.
- Saraf, S., Shringi, D., and Varadarajan, N. Reconstruction of depth-3 arithmetic circuits with constant top fan-in. In *ECCC*, 2025.
- Schwartz, J. T. Fast probabilistic algorithms for verification of polynomial identities. *Journal of the ACM (JACM)*, 27(4):701–717, 1980.
- Tomozei, D.-C. and Massoulié, L. Distributed user profiling via spectral methods. *Stochastic Systems*, 4(1): 1–43, 2014.
- Vakil, R. The rising sea: Foundations of algebraic geometry. 2025.
- Vergari, A., Choi, Y., Liu, A., Teso, S., and den Broeck, G. V. A compositional atlas of tractable circuit operations for probabilistic inference. In *Advances in Neural Information Processing Systems 34: Annual Conference on Neural Information Processing Systems 2021, NeurIPS 2021, December 6-14, 2021, virtual*, pp. 13189–13201, 2021.
- Wedenig, T., Nagpal, R., Cassiers, G., Mangard, S., and Peharz, R. Exact soft analytical side-channel attacks using tractable circuits. In Salakhutdinov, R., Kolter, Z., Heller, K., Weller, A., Oliver, N., Scarlett, J., and Berkenkamp, F. (eds.), *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 52472–52483. PMLR, 21–27 Jul 2024. URL <https://proceedings.mlr.press/v235/wedenig24a.html>.

- Zhang, H., Juba, B., and Van den Broeck, G. Probabilistic generating circuits. In *International conference on machine learning*, pp. 12447–12457. PMLR, 2021.
- Zhang, H., Kung, P.-N., Yoshida, M., Van den Broeck, G., and Peng, N. Adaptable logical control for large language models. *Advances in Neural Information Processing Systems*, 37:115563–115587, 2024.
- Zhang, H., Dang, M., Wang, B., Ermon, S., Peng, N., and Van Den Broeck, G. Scaling probabilistic circuits via monarch matrices. In Singh, A., Fazel, M., Hsu, D., Lacoste-Julien, S., Berkenkamp, F., Maharaj, T., Wagstaff, K., and Zhu, J. (eds.), *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 74804–74818. PMLR, 13–19 Jul 2025. URL <https://proceedings.mlr.press/v267/zhang25q.html>.
- Zippel, R. Probabilistic algorithms for sparse polynomials. In *International symposium on symbolic and algebraic manipulation*, pp. 216–226. Springer, 1979.

A. Pre-processing: Identifying common factors

Since we have oracle access to the probability generating polynomial (PGP), we will rely on standard tools from algebraic complexity for our algorithm. In particular, we use Polynomial Identity Testing (PIT).

A.1. Polynomial Identity Testing (PIT)

The PIT problem asks whether a given polynomial is identically zero. Explicitly expanding a multivariate polynomial may require inspecting exponentially many monomials, but PIT can be performed efficiently via randomized evaluation: a nonzero polynomial remains nonzero under a random substitution with high probability.

Lemma A.1 (Zippel 1979, Schwartz 1980). *Let $p(x_1, \dots, x_n)$ be a nonzero polynomial of total degree at most d over a field $\mathbb{K}$. Let $S \subseteq \mathbb{K}$ be a finite set, and sample $a_1, \dots, a_n \in S$ uniformly at random. Then*

$$\Pr[p(a_1, \dots, a_n) \neq 0] \geq 1 - \frac{d}{|S|}.$$

A.2. Preprocessing step

Given f as in Equation 1, in this step, we reduce the effective number of variables by (i) removing variables that do not appear in f , and (ii) factoring out any linear term common to all mixture components.

Variables not appearing in f . A variable x_j does not appear in f if and only if $f(1, \dots, 1) = f(1, \dots, 1, 0, 1, \dots, 1)$, where the 0 occurs in the j -th position. This is because, using Equation 1 and Remark 1.1, $f(1, \dots, 1, 0, 1, \dots, 1) = \mu_i p_{ij} < \sum_i \mu_i = f(1, \dots, 1)$. Let $S_{\text{not}} \subseteq [n]$ be the set of such indices. This can be found using $1 + n$ queries: first query $f(1, \dots, 1)$ once, and then query f with $x_j = 0$ and all other variables set to 1 for each j .

Common linear factors. Fix $j \notin S_{\text{not}}$ and define the univariate polynomial

$$g_j(x) := f(1, \dots, 1, x, 1, \dots, 1).$$

By Equation 1, g_j is affine in x and $g_j(1) = f(1, \dots, 1) \neq 0$. Set

$$p_j := \frac{g_j(0)}{g_j(1)} \quad \text{and} \quad z_j := -\frac{p_j}{1 - p_j}.$$

Then $(1 - p_j)x_j + p_j$ divides f if and only if the polynomial obtained by substituting $x_j = z_j$ is identically zero:

$$f(x_1, \dots, x_{j-1}, z_j, x_{j+1}, \dots, x_n) \equiv 0.$$

We can test this condition using PIT (Lemma A.1). Let $S_f \subseteq [n]$ denote the set of indices for which this test succeeds, and store the corresponding parameters p_j for each $j \in S_f$.

Overall, this preprocessing uses $O(n)$ oracle evaluations and identifies all such common linear factors along with their exact parameters. Henceforth, whenever we query the PGP, we will set all variables in

$$S_b := S_f \cup S_{\text{not}}$$

to 1 and work with the resulting reduced polynomial on $[n] \setminus S_b$.

The details have been summarized in Algorithm 4. In the PIT check in the algorithm, we view $f(x_1, \dots, x_{j-1}, z_j, x_{j+1}, \dots, x_n)$ as a polynomial in the remaining variables and test whether it is identically zero using Lemma A.1, which is just evaluation of the polynomial at some random points, which is $O(1)$ in the oracle access model, and $O(s)$ in the general setting, where s is the size of the PGC.

It is easy to see that we only make $O(n)$ queries to the PGP in the algorithm. Further, in each iteration, we do at most one PIT check. Hence, the query complexity is $O(n)$. The running time is $O(n)$ in oracle access model, and $O(ns)$ in the general case.

Algorithm 4 Preprocessing: removing irrelevant variables and identifying all common factors

```

1: Input: Oracle access to  $f(x_1, \dots, x_n)$ 
2: Output:  $S_{\text{not}} \subseteq [n]$ ,  $S_f \subseteq [n]$ , and parameters  $\{p_j\}_{j \in S_f}$ 
3: Query  $F \leftarrow f(1, 1, \dots, 1)$ 
4:  $S_{\text{not}} \leftarrow \emptyset$ ,  $S_f \leftarrow \emptyset$ 
5: for  $j = 1$  to  $n$  do
6:   Query  $F_j \leftarrow f(1, \dots, 1, \underbrace{0}_{j\text{-th}}, 1, \dots, 1)$ 
7:   if  $F_j = F$  then
8:      $S_{\text{not}} \leftarrow S_{\text{not}} \cup \{j\}$   $\{x_j \text{ does not appear in } f\}$ 
9:   else
10:     $p_j \leftarrow F_j / F$ 
11:     $z_j \leftarrow -p_j / (1 - p_j)$ 
12:    if  $f(x_1, \dots, x_{j-1}, z_j, x_{j+1}, \dots, x_n) \equiv 0$  (via Lemma A.1) then
13:       $S_f \leftarrow S_f \cup \{j\}$ 
14:      Store parameter  $p_j$ 
15:    end if
16:  end if
17: end for
    
```

B. Jacobian and symbolic rank

Let $\Phi = (\Phi_1, \dots, \Phi_k) : \mathbb{K}^d \rightarrow \mathbb{K}^k$ be a polynomial map over a field $\mathbb{K}$, and let $\theta = (\theta_1, \dots, \theta_d) \in \mathbb{K}^d$ denote the parameter vector (in our setting, θ collects parameters such as μ_i and t_{ij}). The *Jacobian* of Φ is the $k \times d$ matrix

$$J_\Phi(\theta) := \left(\frac{\partial \Phi_a}{\partial \theta_b} \right)_{a \in [k], b \in [d]}, \quad (7)$$

which we view as a *symbolic* matrix with entries in the polynomial ring $\mathbb{K}[\theta]$. Its *symbolic rank* is the rank of $J_\Phi(\theta)$ over the function field $\mathbb{K}(\theta)$, equivalently the largest t such that some $t \times t$ minor is not the zero polynomial. In particular, if $J_\Phi(\theta)$ has full column rank symbolically, then it has full column rank for all parameter values outside a proper Zariski-closed set, yielding a generic non-degeneracy condition (Vakil, 2025; Leung et al., 2016). When $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, full column rank at a point further implies local injectivity of Φ in a neighborhood of that point by the inverse/implicit function theorem (Lee, 2012).

Remark B.1. In our setting, we first apply a restriction to project the PGP to $m = O(r)$ variables, and then view the resulting low-degree coefficients (e.g., all degree ≤ 3 coefficients) as polynomial functions of the underlying parameters θ . This defines a polynomial map Φ from parameters to observable coefficients. We analyze the symbolic Jacobian of this map to show that it has *large* rank generically, which in turn implies that the restricted instance remains identifiable for almost all parameter choices.

C. Rank bound on the Jacobian

In this section we show how we can bound the rank of the Jacobian matrix J_r formed from the distribution coefficients of a r -mixture. We demonstrate that there is a separation in these ranks for any $r_1 \neq r_2$. For ease of reading, we begin with the special case $r = 2$, which serves as a gentle introduction to the main ideas.

C.1. Toy case: separating $r = 2$ from $r = 1$

In this subsection we illustrate the main idea behind our rank-based separation argument by analyzing the smallest nontrivial case: distinguishing fan-in 2 from fan-in 1. The general rank bound for arbitrary r is proved later.

W.l.o.g. let $A = [m]$. For $r = 1$, the parameters are $\mu, t_1, \dots, t_m$ (for simplicity), and the Jacobian J_1 (with rows indexed by $S \subseteq [m]$ and columns indexed by $\{\mu\} \cup \{t_j\}_{j \in [m]}$) has 2^m rows and $m + 1$ columns. In particular, $\text{rank}(J_1) \leq m + 1$, and in fact, it is easy to show that $\text{rank}(J_1) = m + 1$. Consider the $(m + 1) \times (m + 1)$ submatrix of J_1 obtained by taking rows indexed by $S = \emptyset, \{1\}, \dots, \{m\}$ and columns indexed by $\mu, t_1, \dots, t_m$. This submatrix is upper triangular with diagonal entries $1, \mu, \dots, \mu$, and hence has determinant $\mu^m \neq 0$ as a polynomial.

Fact 1. The rank of J_1 is $m + 1$.

For $r = 2$, the Jacobian J_2 has 2^m rows and $2(m + 1)$ columns. It has the following form:

$$\begin{array}{c}
 c_{\emptyset} \\
 c_{\{1\}} \\
 \vdots \\
 c_{\{1,2\}} \\
 \vdots \\
 c_{[m]}
 \end{array}
 \begin{pmatrix}
 \mu_1 & \mu_2 & t_{11}, \dots, t_{1m} & t_{21}, \dots, t_{2m} \\
 1 & 1 & 0, \dots, 0 & 0, \dots, 0 \\
 * & * & v_{1,\{1\}} & v_{2,\{1\}} \\
 \vdots & \vdots & \vdots & \vdots \\
 * & * & v_{1,\{1,2\}} & v_{2,\{1,2\}} \\
 \vdots & \vdots & \vdots & \vdots \\
 * & * & v_{1,[m]} & v_{2,[m]}
 \end{pmatrix} \quad (8)$$

where $v_{k,S}$ denotes the row vector corresponding to derivatives of c_S with respect to the variables $\{t_{k1}, \dots, t_{km}\}$. For example, for $S = \{1, 2\}$ we have $c_S = \mu_1 t_{11} t_{12} + \mu_2 t_{21} t_{22}$, and hence $v_{1,S} = [\mu_1 t_{12} \ \mu_1 t_{11} \ 0 \ \dots \ 0]$.

We now isolate the contribution of the constant coefficient row $c_{\emptyset}$. Using row operations, we can make all entries in the first μ_1 -column zero except in the first row. Next, using column operations among the μ_1, μ_2 columns, we can make the top entry of the μ_2 -column equal to 0. Since all t -columns have a 0 in the first row, these operations do not affect any $v_{k,S}$ block. As a result, the rank of J_2 is at least *one more* than the rank of the submatrix obtained by restricting to: (i) rows indexed by nonempty sets $S \neq \emptyset$, and (ii) columns indexed only by the t_{ij} 's.

Proposition C.1. *Let M be the submatrix of J_2 whose rows are indexed by nonempty $S \subseteq [m]$ and whose columns are indexed by $\{t_{ij}\}_{i \in [2], j \in [m]}$. Then*

$$\text{rank}(J_2) \geq 1 + \text{rank}(M).$$

We now lower bound $\text{rank}(M)$. Group the rows of M according to $|S|$, and view M in the following block form:

$$M = \begin{array}{c} |S|=1 \\ |S|=2 \\ \vdots \\ |S|=m \end{array} \begin{pmatrix} (t_{1j})_{j \in [m]} & (t_{2j})_{j \in [m]} \\ \mu_1 I_m & \mu_2 I_m \\ \mu_1 B_{1,2} & \mu_2 B_{2,2} \\ \vdots & \vdots \\ \mu_1 B_{1,m} & \mu_2 B_{2,m} \end{pmatrix}$$

where $B_{i,k}$ is a $\binom{m}{k} \times m$ matrix whose entries are polynomials in $\{t_{i,j}\}_{j \in [m]}$. Concretely, the rows of $B_{i,k}$ are the vectors $\frac{1}{\mu_i} v_{i,S}$ over all k -subsets $S \subseteq [m]$.

Since we only care about rank (and work over a field), we may apply row and column operations; in particular, we may assume $\mu_1 \neq 0$ (otherwise swap the two mixture components). Perform the following column operations: for each $j \in [m]$, replace the column t_{2j} by $t_{2j} \leftarrow t_{2j} - \frac{\mu_2}{\mu_1} t_{1j}$. This makes the singleton block in the (t_{2j}) columns identically zero, while the $\mu_1 I_m$ block in the (t_{1j}) columns remains unchanged. Hence $\text{rank}(M) \geq m$.

Next consider any doubleton row $S = \{a, b\}$. Before the column operation, the entry in this row and the column t_{2a} equals $\mu_2 t_{2b}$, while the entry in the same row and the column t_{1a} equals $\mu_1 t_{1b}$. After the column operation, the new entry in column t_{2a} becomes $\mu_2 t_{2b} - \frac{\mu_2}{\mu_1} (\mu_1 t_{1b}) = \mu_2 (t_{2b} - t_{1b})$, which is a nonzero polynomial. Therefore, the transformed (t_{2j}) block contains a nonzero entry in some doubleton row, while all singleton rows have zeros in the (t_{2j}) block. This yields at least one additional independent direction beyond the m pivots coming from the $\mu_1 I_m$ block, implying $\text{rank}(M) \geq m + 1$. Combining with Proposition C.1, we conclude that $\text{rank}(J_2) \geq m + 2$, separating the $r = 2$ case from the $r = 1$ case.

Fact 2. The rank of J_2 is at least $m + 2$.

C.2. Rank bound on the Jacobian in the general setting

We now prove rank-separation in the general r case. Recall that the Jacobian J_r has rows indexed by $S \subseteq [m]$ and columns indexed by the parameters $\{\mu_i\}_{i \in [r]} \cup \{t_{i\ell}\}_{i \in [r], \ell \in [m]}$. Let M denote the submatrix of J_r obtained by

columns indexed by $\{t_{i\ell}\}_{i \in [r], \ell \in [m]}$, rows indexed by S (except $S = \emptyset$). Thus M has size $2^m - 1 \times mr$ and entries

$$M_{S,(i,\ell)} = \frac{\partial c_S}{\partial t_{i\ell}} = \mu_i \cdot \mathbf{1}_{\{\ell \in S\}} \prod_{j \in S \setminus \{\ell\}} t_{ij}.$$

In J_r , the μ -columns contribute at most r additional rank, while the row $S = \emptyset$ contributes at least one rank (because $\partial c_\emptyset / \partial \mu_i = 1$ for all i and $\partial c_\emptyset / \partial t_{i\ell} = 0$). Therefore,

$$1 + \text{rank}(M) \leq \text{rank}(J_r) \leq r + \text{rank}(M). \quad (9)$$

Hence, it suffices to determine $\text{rank}(M)$.

A specialization. To lower bound the *symbolic* rank of M , it is enough to exhibit one assignment of parameters for which M attains large rank. We specialize to

$$t_{ij} = a_i \cdot b_j \quad \text{for all } i \in [r], j \in [m], \quad (10)$$

where $a_i, b_j \neq 0$. (Since we only use this as a witness assignment for rank, the signs are irrelevant; in our model one may take $a_i, b_j > 0$.)

Under (10), for any $S \subseteq [m]$ with $|S| = k$ and any $\ell \in [m]$,

$$\frac{\partial c_S}{\partial t_{i\ell}} = \mu_i \cdot \mathbf{1}_{\{\ell \in S\}} \prod_{j \in S \setminus \{\ell\}} (a_i b_j) = \mu_i a_i^{k-1} \cdot \mathbf{1}_{\{\ell \in S\}} \prod_{j \in S \setminus \{\ell\}} b_j.$$

Grouping the rows of M by $k = |S|$, we can write M in the following block form:

$$M = \begin{matrix} & \begin{matrix} (t_{11}, \dots, t_{1m}) & (t_{21}, \dots, t_{2m}) & \cdots & (t_{r1}, \dots, t_{rm}) \end{matrix} \\ \begin{matrix} |S|=1 \\ |S|=2 \\ \vdots \\ |S|=k \\ \vdots \\ |S|=m \end{matrix} & \begin{pmatrix} \mu_1 I_m & \mu_2 I_m & \cdots & \mu_r I_m \\ \mu_1 a_1 G_2 & \mu_2 a_2 G_2 & \cdots & \mu_r a_r G_2 \\ \vdots & \vdots & \ddots & \vdots \\ \mu_1 a_1^{k-1} G_k & \mu_2 a_2^{k-1} G_k & \cdots & \mu_r a_r^{k-1} G_k \\ \vdots & \vdots & \ddots & \vdots \\ \mu_1 a_1^{m-1} G_m & \mu_2 a_2^{m-1} G_m & \cdots & \mu_r a_r^{m-1} G_m \end{pmatrix} \end{matrix} \quad (11)$$

where for each $k \in \{1, \dots, m\}$ the matrix G_k is a $\binom{m}{k} \times m$ matrix depending only on $(b_1, \dots, b_m)$, with rows indexed by k -subsets S and columns indexed by $\ell \in [m]$, and entry

$$(G_k)_{S,\ell} = \mathbf{1}_{\{\ell \in S\}} \prod_{j \in S \setminus \{\ell\}} b_j.$$

Note that $G_1 = I_m$, and G_m is a $1 \times m$ row vector with all entries nonzero.

Lemma C.2. *For every $k \in \{1, 2, \dots, m-1\}$, we have $\text{rank}(G_k) = m$.*

Proof. We prove the claim by exhibiting one assignment of $(b_1, \dots, b_m)$ for which G_k has full column rank. Set $b_1 = \dots = b_m = 1$. Then $(G_k)_{S,\ell} = \mathbf{1}_{\{\ell \in S\}}$, i.e., G_k becomes the incidence matrix of k -subsets vs. elements.

Suppose $\alpha \in \mathbb{K}^m$ satisfies $G_k \alpha = 0$. Then for every k -subset $S \subseteq [m]$ we have $\sum_{\ell \in S} \alpha_\ell = 0$. Fix $i \neq j$ and choose a set $T \subseteq [m] \setminus \{i, j\}$ of size $k-1$ (possible since $k \leq m-1$). Apply the constraint to $S = T \cup \{i\}$ and $S' = T \cup \{j\}$ and subtract to obtain $\alpha_i - \alpha_j = 0$. Thus all coordinates of α are equal: $\alpha_1 = \dots = \alpha_m = \alpha$. Plugging into any constraint gives $k\alpha = 0$, hence $\alpha = 0$ (over characteristic 0, e.g. $\mathbb{R}$ or $\mathbb{C}$). Therefore the nullspace is trivial and $\text{rank}(G_k) = m$. $\square$

Reducing to a Vandermonde structure. By Lemma C.2, for each $k \in \{1, \dots, m-1\}$ there exists an invertible matrix of row operations R_k such that $R_k G_k$ has the form $\begin{bmatrix} I_m \\ 0 \end{bmatrix}$ (i.e., its first m rows form I_m). Applying these row operations independently within each $|S| = k$ block of (11) (and discarding the resulting all-zero rows), we obtain a submatrix M' of M with $(m-1) \cdot m + 1$ rows and rm columns of the form

$$M' = \begin{matrix} & (t_{11}, \dots, t_{1m}) & (t_{21}, \dots, t_{2m}) & \cdots & (t_{r1}, \dots, t_{rm}) \\ \begin{matrix} k=1 \\ k=2 \\ \vdots \\ k=m-1 \\ k=m \end{matrix} & \begin{pmatrix} \mu_1 I_m & \mu_2 I_m & \cdots & \mu_r I_m \\ \mu_1 a_1 I_m & \mu_2 a_2 I_m & \cdots & \mu_r a_r I_m \\ \vdots & \vdots & \ddots & \vdots \\ \mu_1 a_1^{m-2} I_m & \mu_2 a_2^{m-2} I_m & \cdots & \mu_r a_r^{m-2} I_m \\ \mu_1 a_1^{m-1} \tilde{u} & \mu_2 a_2^{m-1} \tilde{u} & \cdots & \mu_r a_r^{m-1} \tilde{u} \end{pmatrix} \end{matrix} \quad (12)$$

where $\tilde{u}$ is a fixed nonzero $m \times 1$ vector (coming from G_m). Clearly $\text{rank}(M) = \text{rank}(M')$ and hence $\text{rank}(M) \leq \min\{rm, (m-1)m + 1\}$.

Now assume $m \geq r+1$ and choose $\mu_i \neq 0$ and pairwise distinct $a_1, \dots, a_r$. Consider the submatrix of M' formed by taking the first r block-rows (i.e., $k = 1, 2, \dots, r$) and all columns. This submatrix equals

$$\left(\begin{bmatrix} 1 & 1 & \cdots & 1 \\ a_1 & a_2 & \cdots & a_r \\ \vdots & \vdots & \ddots & \vdots \\ a_1^{r-1} & a_2^{r-1} & \cdots & a_r^{r-1} \end{bmatrix} \otimes I_m \right) \cdot \text{diag}(\mu_1, \dots, \mu_r),$$

which has rank rm since the $r \times r$ Vandermonde matrix is nonsingular when the a_i 's are distinct. Therefore $\text{rank}(M') = rm$, and hence $\text{rank}(M) = rm$ for this assignment. Consequently, the symbolic rank of M is rm whenever $m \geq r+1$.

Combining with (3), for $m \geq r+1$ we obtain

$$rm + 1 \leq \text{rank}(J_r) \leq rm + r. \quad (13)$$

Remark C.3. The lower bound above fails only on a proper algebraic subset of the parameter space (where all $rm \times rm$ minors of M vanish). Hence, in the generic setting and for $m \geq r+1$, the ranks of the Jacobians separate different mixture sizes. In particular, this separation holds for arbitrary choices of the weights $\{\mu_i\}$ (as long as $\mu_i \neq 0$), since μ only appears as a nonzero diagonal scaling in the Vandermonde witness argument above.

D. Interpolation

Given query access to a polynomial f , a basic task is to recover its low-degree coefficients. By Equation 1, our PGP is multilinear (i.e., each variable appears with exponent either 0 or 1 in any monomial). In our algorithm we will need to extract all degree-2 and degree-3 coefficients.

Lemma D.1. Suppose we have query access to a multilinear polynomial $f \in \mathbb{C}[x_1, \dots, x_n]$ of degree at most n :

$$f(x_1, \dots, x_n) = \sum_{S \subseteq [n]} c_S \prod_{i \in S} x_i.$$

Then we can recover c_S for all $|S| \leq 3$ using $O(n^3)$ queries.

Proof. For each $S \subseteq [n]$, let $e_S \in \{0, 1\}^n$ denote the indicator vector of S , i.e., $e_S(j) = 1$ if $j \in S$ and $e_S(j) = 0$ otherwise. Since f is multilinear,

$$f(e_S) = \sum_{T \subseteq S} c_T.$$

In particular, $c_\emptyset = f(e_\emptyset) = f(0, \dots, 0)$. For each $i \in [n]$,

$$c_{\{i\}} = f(e_{\{i\}}) - c_\emptyset.$$

For each distinct $i, j \in [n]$,

$$c_{\{i,j\}} = f(e_{\{i,j\}}) - c_{\{i\}} - c_{\{j\}} + c_\emptyset.$$

For each distinct $i, j, k \in [n]$,

$$c_{\{i,j,k\}} = f(e_{\{i,j,k\}}) - c_{\{i,j\}} - c_{\{i,k\}} - c_{\{j,k\}} - c_{\{i\}} - c_{\{j\}} - c_{\{k\}} - c_\emptyset.$$

Thus, by querying $f(e_S)$ for all S of size at most 3 and storing these values, we can compute all coefficients c_S with $|S| \leq 3$. The number of such subsets is $1 + \binom{n}{1} + \binom{n}{2} + \binom{n}{3} = O(n^3)$. $\square$

Remark D.2. We will only apply Lemma D.1 after restricting f to $m = O(r)$ variables, so the cost becomes $O(m^3) = \text{poly}(r)$ per restriction.

E. Recovering missing entries of M_2 and M_3 from off-diagonal entries

We show here how, in the generic setting, we can recover the missing diagonal (and repeated-index) entries of M_2 and M_3 using only the off-diagonal entries obtained via interpolation. Formally, we show the following.

Proposition E.1. *Let M be a real symmetric matrix of size $m \times m$ and rank r , and T be a real third-order symmetric tensor of size $m \times m \times m$ and CP rank r .*

1. *Suppose we have access to all off-diagonal entries of M , ie, we are given $M_{i,j}$ for all $i \neq j$ where $i, j \in [m]$. Then, in the generic setting, we can recover all the diagonal entries of M ($M_{i,i}$ for $i \in [m]$) in time $\tilde{O}(m^2 + mr^{1+\omega})$ where ω is the matrix multiplication exponent.*
2. *Suppose we have access to all off-diagonal entries of T , ie, we are given $T_{i,j,k}$ for all pairwise distinct $i, j, k \in [m]$. Then, in the generic setting, we can recover all the diagonal and repeated index entries of T ($T_{i,j,k}$ for $i, j, k \in [m]$ where at least two of the three indices are equal) in time in time $\tilde{O}(m^3 + m^2r^{1+\omega})$.*

Further, these entries are unique in the generic setting. Here, $\tilde{O}(\cdot)$ denotes standard complexity variant of $O(\cdot)$ which subsumes polylogarithmic factors.

The proof of the proposition follows in the subsequent subsections.

E.1. Vanishing of minors of a symmetric rank- r matrix in the generic setting

Lemma E.2. *Let A be a real symmetric $n \times n$ matrix of rank exactly r . Then, generically over the set of rank- r symmetric matrices, all $r \times r$ minors of A are non-zero.*

Proof. Let S_n denote the set of all $n \times n$ real symmetric matrices. For $I, J \subseteq [n]$ with $|I| = |J|$, define

$$m_{I,J}(A) := \det(A_{I,J}),$$

where $A_{I,J}$ denotes the submatrix of A with rows indexed by I and columns indexed by J . Clearly, $m_{I,J}$ is a polynomial function on S_n .

Let

$$\mathcal{R}_r := \{A \in S_n : \text{rank}(A) = r\}.$$

Fix I, J with $|I| = |J| = r$. We will show that $m_{I,J}$ is not identically zero on $\mathcal{R}_r$.

Consider a Vandermonde-type matrix $V \in \mathbb{R}^{n \times r}$ defined by choosing distinct nonzero scalars $t_1, \dots, t_n$ and setting

$$V_{p,q} = t_p^{q-1}.$$

For any $I \subseteq [n]$ with $|I| = r$, the submatrix V_I (rows indexed by I) is a Vandermonde matrix on distinct parameters and hence satisfies $\det(V_I) \neq 0$.

Now define $M := VV^\top$. Clearly $M \in S_n$. Since V has full column rank r , we have

$$\text{rank}(M) = \text{rank}(VV^\top) = \text{rank}(V) = r,$$

and hence $M \in \mathcal{R}_r$.

Moreover, for any $I, J \subseteq [n]$ with $|I| = |J| = r$, we have

$$M_{I,J} = V_I V_J^\top.$$

Therefore,

$$\det(M_{I,J}) = \det(V_I) \cdot \det(V_J^\top) = \det(V_I) \cdot \det(V_J) \neq 0.$$

Hence $m_{I,J}(M) \neq 0$, and thus $m_{I,J}$ is not identically zero on $\mathcal{R}_r$.

For each (I, J) , define the zero set

$$Z_{I,J} := \{A \in \mathcal{R}_r : m_{I,J}(A) = 0\}.$$

Since $m_{I,J}$ is a polynomial and not identically zero on $\mathcal{R}_r$, the set $Z_{I,J}$ is a proper algebraic subset of $\mathcal{R}_r$. Let

$$Z := \bigcup_{|I|=|J|=r} Z_{I,J}.$$

This is a finite union of proper algebraic subsets, so Z is still a proper algebraic subset of $\mathcal{R}_r$. Since $M \notin Z$, it follows that outside Z (i.e., generically on $\mathcal{R}_r$), all $r \times r$ minors are non-zero. $\square$

This lemma is useful for reconstructing the missing diagonal entries of M_2 from its off-diagonal entries. Fix a restriction of size $m \geq 2r + 1$ and obtain all off-diagonal entries of M_2 (e.g., by interpolation on degree-2 coefficients). Now choose two disjoint r -subsets $A_1, A_2 \subseteq [m]$ such that $A_1 \cap A_2 = \emptyset$, and pick an index

$$\ell \in [m] \setminus (A_1 \cup A_2).$$

Consider the $(r+1) \times (r+1)$ submatrix K of M_2 whose rows are indexed by $A_1 \cup \{\ell\}$ and whose columns are indexed by $A_2 \cup \{\ell\}$. By construction, K contains exactly one diagonal entry of M_2 , namely

$$x := (M_2)_{\ell,\ell},$$

and all other entries of K are off-diagonal and hence already known.

Since $\text{rank}(M_2) = r$, every $(r+1) \times (r+1)$ minor vanishes, so $\det(K) = 0$. Expanding $\det(K)$ along the row/column containing x , we obtain an identity of the form

$$x C_1 + C_2 = 0,$$

where C_1 is an $r \times r$ minor of M_2 (specifically, the cofactor corresponding to x), and C_2 depends only on the known off-diagonal entries. By Lemma E.2, generically $C_1 \neq 0$, and hence we can solve uniquely for

$$x = -\frac{C_2}{C_1}.$$

To compute x numerically, it suffices to compute C_1 (a determinant of size r) and C_2 (which can be obtained, for example, by evaluating $\det(K)$ after setting $x = 0$, i.e., one determinant of size $r+1$). Thus, the time to recover a single diagonal entry is $T_r + T_{r+1}$, where T_k denotes the time to compute the determinant of a $k \times k$ numerical matrix. Using fast linear algebra, $T_k = O(k^\omega)$. Further, if all the parameters had precision upto ϵ , then each off-diagonal entry of M_2 can have precision upto ϵ^2 . The determinant of any such matrix will have a precision

of at most $O(r^{\frac{r}{2}} 2^{r \log \frac{1}{\epsilon}})$ (Bareiss, 1972). Hence, we need to compute it upto $O(r(\log r + \log \frac{1}{\epsilon}))$ bits to compute the diagonal entries. Hence, recovering all m diagonal entries takes total time $O(mr^{1+\omega}(\log r + \log \frac{1}{\epsilon}))$.

Using Lemma D.1, we only need $O(m^2)$ queries to obtain all off-diagonal entries of M_2 . In case of oracle access to the PGP this leads to time complexity of $O(m^2)$, while in the general case it is $O(sm^2)$, where s is the size of the PGC. Combining this with the time to reconstruct the diagonal entries, the total time to recover the full *exact* matrix M_2 is

$$\tilde{O}(m^2 + mr^{1+\omega})$$

in the oracle access model, and $\tilde{O}(sm^2 + mr^{1+\omega})$ in the general case. Further, the query complexity is $O(m^2)$ for a fixed alive set.

E.2. Vanishing of minors of a third-order rank r tensor in the generic setting

In M_2 , interpolation helps us recover all off-diagonal entries. For M_3 as above, we can again recover all entries $(M_3)_{a,b,c}$ for all *pairwise distinct* a, b, c by interpolation. However, we are again missing entries where $a = b = c$, or where exactly two of the three indices coincide. We show that using similar ideas as above, we can recover these entries *exactly*.

Let $T \in \mathbb{R}^{n \times n \times n}$ be a third-order symmetric tensor of the form

$$T = \sum_{i=1}^r a_i u_i \otimes u_i \otimes u_i, \quad a_i \neq 0, \quad (14)$$

where $u_1, \dots, u_r \in \mathbb{R}^n$ are linearly independent. Let $U = [u_1 \dots u_r] \in \mathbb{R}^{n \times r}$ and $\Lambda = \text{diag}(a_1, \dots, a_r)$. Using the matricized CP identity (Kolda & Bader, 2009, Equation (3.2)), the mode-1 unfolding of T can be written as

$$T_{(1)} = U \Lambda (U \odot U)^\top,$$

where $\odot$ denotes the Khatri–Rao product (columnwise Kronecker product). Note that $(U \odot U) \in \mathbb{R}^{n^2 \times r}$ is precisely the matrix

$$L = [u_1 \otimes u_1 \quad u_2 \otimes u_2 \quad \dots \quad u_r \otimes u_r].$$

In particular, L has at most r columns, and in the generic setting these columns are linearly independent, i.e., $\text{rank}(L) = r$.² Since $\text{rank}(U) = r$ and $\text{rank}(\Lambda) = r$, we obtain

$$\text{rank}(T_{(1)}) = \text{rank}(U \Lambda L^\top) = r.$$

By symmetry of T , the same argument gives $\text{rank}(T_{(2)}) = \text{rank}(T_{(3)}) = r$. Moreover, from (14) we have $\text{rank}_{\text{CP}}(T) \leq r$. Since every flattening rank is a lower bound on CP rank (Christandl & Zuiddam, 2019, Section 2.3), it follows that $\text{rank}_{\text{CP}}(T) = r$.

We now use these rank constraints to recover the missing entries of M_3 . Fix a restriction of size m and let $F := T_{(1)}$ denote the mode-1 unfolding, viewed as a matrix of size $m \times m^2$. Index the columns of F by ordered pairs $(j, k) \in [m] \times [m]$, so that

$$F_{i, (j, k)} = T_{i, j, k}.$$

From the above discussion, in the generic setting we have $\text{rank}(F) = r$, and hence every $(r+1) \times (r+1)$ minor of F vanishes.

Recovering entries of the form $T_{i, i, k}$ with $i \neq k$. We choose an $(r+1) \times (r+1)$ submatrix of F that contains *exactly one* unknown entry of this form. Concretely, pick index sets

$$R \subseteq [m] \text{ with } |R| = r+1, \quad C \subseteq [m] \times [m] \text{ with } |C| = r+1,$$

such that: (i) for all but one column $(j, k) \in C$, we have j, k distinct from every $i \in R$, so that the corresponding entries $T_{i, j, k}$ are already known from interpolation (pairwise distinct indices), and (ii) there is exactly one special

²For example, taking U to be a Vandermonde-type matrix as in the proof of Lemma E.2 yields $\text{rank}(L) = r$, and hence $\text{rank}(L) = r$ holds generically.

column $(i^*, k^*) \in C$ with $i^* \in R$ and $k^* \notin R$, for which the unique unknown entry is T_{i^*, i^*, k^*} (two indices equal). Let B denote the resulting $(r+1) \times (r+1)$ submatrix of F with rows R and columns C . Then $\det(B) = 0$ since $\text{rank}(F) = r$. Moreover, $\det(B)$ is *linear* in the single unknown entry T_{i^*, i^*, k^*} (all other entries are known), and expanding along the position of this unknown gives an equation of the form

$$T_{i^*, i^*, k^*} \cdot C_1 + C_2 = 0,$$

where C_1 is an $r \times r$ minor of F and C_2 depends only on known entries. By a genericity argument identical to Lemma E.2 (applied to the rank- r matrix F) and accounting for precision as above (here each entry of T can have precision upto ϵ^3), we have $C_1 \neq 0$ generically, and hence we can solve for T_{i^*, i^*, k^*} exactly.

There are $\Theta(m^2)$ such entries (up to symmetry), so all entries with exactly two equal indices can be recovered in time $\tilde{O}(m^2 r^{1+\omega})$, in the oracle access model.

Recovering entries of the form $T_{i,i,i}$. Similarly, we recover diagonal entries $T_{i,i,i}$ by selecting a minor containing exactly one entry with all three indices equal, while all other entries correspond to pairwise distinct indices. Since there are only m such values, this takes time $\tilde{O}(mr^{1+\omega})$ after accounting for precision.

Putting the two cases together, the total time (in the oracle access model) to reconstruct all missing entries of M_3 exactly is

$$\tilde{O}(m^2 r^{1+\omega}).$$

Adding the time $O(m^3)$ required to recover all entries with pairwise distinct indices via interpolation, we can recover the full exact tensor M_3 (in the oracle access model) in time

$$\tilde{O}(m^3 + m^2 r^{1+\omega}).$$

and in time $\tilde{O}(sm^3 + m^2 r^{1+\omega})$ in the general setting. Further, the query complexity was only $O(m^3)$.

Combining with the discussion on recovering M_2 , the total query complexity is $O(m^3)$ for each alive set. The running time is: $\tilde{O}(m^3 + m^2 r^{1+\omega})$ in the oracle access model, and $\tilde{O}(sm^3 + m^2 r^{1+\omega})$. Using Proposition 3.1, we only need $m \geq r+1$ to preserve the fan-in r . Further, to reconstruct all missing entries of M_2 and M_3 using the above minor-vanishing procedure, it suffices to take $m \geq 3r+2$. Thus, we can run our algorithm with restrictions of size $m = 3r+2$. Since $m = O(r)$, this gives a running time of $\tilde{O}(r^{3+\omega})$ (in the oracle access model) for recovering M_2 and M_3 exactly on a restriction.

Remark E.3. Our main speedup comes from exploiting Proposition 3.1. If we apply the method of moments naively on n variables, then reconstructing M_2 and M_3 exactly would take $O(n^3)$ queries and a running time of $\tilde{O}(n^3 + n^2 r^{1+\omega})$ in the oracle access model.

F. Speeding up the whitening process for M_3

We saw that whitening M_3 to get $\hat{M}_3$ requires us to do the following computation:

$$(\hat{M}_3)_{abc} = \sum_{i=1}^m \sum_{j=1}^m \sum_{l=1}^m W_{i,a} W_{j,b} W_{l,c} (M_3)_{ijl}$$

Doing the multiplication naively would lead to a running time of $O(m^3 r^3)$. However, using multiplication along the different flattenings, we can achieve significant speed-up in this step. We demonstrate the method now: In the first step, we compute the first intermediate tensor T_1 of size $r \times m \times m$ using M_3 such that

$$(T_1)_{ajl} = \sum_{i=1}^m W_{i,a} (M_3)_{ijl}$$

We have rm^2 entries and computing each entry takes m arithmetic operations. Hence, total time to compute T_1 is $O(rm^3)$. Once, we have T_1 , we use it to get T_2 , a $r \times r \times m$ tensor. Formally,

$$(T_2)_{abl} = \sum_{j=1}^m W_{j,b} (T_1)_{ajl}$$

Here we have $r^2 m$ entries, and hence time taken is $O(r^2 m^2)$. Finally, we use T_2 to get $\hat{M}_3$.

$$(\hat{M}_3)_{abc} = \sum_{l=1}^m W_{l,c}(T_2)_{abl}$$

Here we have r^3 entries and hence time is $O(r^3 m)$. Hence total time to get $\hat{M}_3$ from M_3 is $O(rm^3 + r^2 m^2 + r^3 m)$.

G. Uniqueness of projections

For a fixed fan-in r , for every alive set A we know the number of variables we need to learn, namely $m := |A|$. Moreover, by Proposition 3.1, as long as $m \geq r + 1$, the projected distribution generically has exactly r mixture components, and hence admits a rank- r moment decomposition for any projection.

However, a priori it is still possible that a particular projection is *not identifiable*: namely, there could exist two different parameter tuples that yield the same projected PGP. Concretely, there may exist $f(x_1, \dots, x_m) = \sum_{i=1}^r \mu_i \prod_{j=1}^m (\alpha_{ij} x_j + 1)$, and $g(x_1, \dots, x_m) = \sum_{i=1}^r \delta_i \prod_{j=1}^m (\beta_{ij} x_j + 1)$, such that $f = g$ as polynomials, even though (μ_i, α_{ij}) and (δ_i, β_{ij}) are not the same up to permutation. Equality of polynomials implies equality of all coefficients, and hence for every $\{k_1, \dots, k_\ell\} \subseteq [m]$ we must have $\sum_{i=1}^r \mu_i \prod_{t=1}^\ell \alpha_{i,k_t} = \sum_{i=1}^r \delta_i \prod_{t=1}^\ell \beta_{i,k_t}$.

We show that such non-uniqueness cannot occur in our (generic) setting: for a generic choice of parameters, the projected moments determine the mixture components uniquely (up to permutation), and therefore the above situation is impossible.

We recall that the *Kruskal rank* k_A of a matrix A is the largest integer k such that every set of k columns of A is linearly independent.

Theorem G.1 (Kruskal uniqueness for tensor decomposition, [Kruskal \(1977\)](#)). *Let $T \in \mathbb{R}^{m \times m \times m}$ admit a rank- r tensor decomposition of the form*

$$T = \sum_{i=1}^r a_i \otimes b_i \otimes c_i,$$

and let $A = [a_1 | \dots | a_r]$, $B = [b_1 | \dots | b_r]$, $C = [c_1 | \dots | c_r]$ be the corresponding factor matrices. Let k_A, k_B, k_C denote their Kruskal ranks. If

$$k_A + k_B + k_C \geq 2r + 2,$$

then the above decomposition is unique up to permutation of the r terms and rescaling within each term.

Since we are looking at tensors of the form Equation 14, it is a special case of Theorem G.1 with factor matrices $A = B = C = U$, where $U = [u_1 | \dots | u_r] \in \mathbb{R}^{m \times r}$. Therefore, uniqueness holds whenever

$$3k_U \geq 2r + 2.$$

For $m \geq r$, the matrix U has Kruskal rank $k_U = r$ for generic choices of $\{u_i\}_{i=1}^r$, and hence the decomposition $T = \sum_{i=1}^r \mu_i u_i^{\otimes 3}$ is generically unique up to permutation and rescaling. Since we also use the second-order moments for reconstruction, we can always avoid the scaling issue simply as the method of moments learns the correct weights μ_i and parameters α_{ij} from given M_2, M_3 .

H. Avoiding collisions in reconstruction

We would first like to point out that suppose we look at a projection under an alive set A as shown in Equation 2. Then, for distinct μ_i , the μ_i s themselves act as the identifier of the particular mixture. However, suppose w.l.o.g. $\mu_1 = \mu_2$. Then, it can't be the case that $\prod_{j \in A} (\alpha_{1j} x_j + 1) = \prod_{j \in A} (\alpha_{2j} x_j + 1)$, as then in the particular projection A , the fan-in would not be r , which contradicts Proposition 3.1.

Proposition H.1. *Suppose $\mu_i = \mu_j$. For projections of size $m \geq r + 1$, the corresponding product distributions can't be identical in the "generic" setting.*

However, we can still run into the following problem: Suppose, our original distribution was parametrized as following: We have (a) $r = 2, n = 4, \mu_1 = \mu_2 = 0.5$, (b) $(\alpha_{11}, \alpha_{12}, \alpha_{13}, \alpha_{14}) = (0.1, 0.2, 0.3, 0.4)$, and (c) $(\alpha_{21}, \alpha_{22}, \alpha_{23}, \alpha_{24}) = (0.6, 0.7, 0.8, 0.9)$.

Let our first alive set be $A_1 = \{1, 2\}$ and second alive set be $A_2 = \{3, 4\}$. Using our reconstruction algorithm with the restriction to A_1 and ignoring permutations, we will recover $\mu_1 = \mu_2 = 0.5$, and $(\alpha_{11}, \alpha_{12}) = (0.1, 0.2)$ and $(\alpha_{21}, \alpha_{22}) = (0.6, 0.7)$. However, when we now take the alive set A_2 , since we only look at $f = \sum_{i=1}^r \mu_i \prod_{j \in A_2} (\alpha_{ij} x_j + 1)$ and $\mu_1 = \mu_2$, we would have no way to distinguish whether $(\alpha_{13}, \alpha_{14})$ takes the values $(0.3, 0.4)$ or $(0.8, 0.9)$. Clearly, if we choose the second tuple, then we will have recovered an incorrect distribution. However, we now show that by only using an overhead of at most $r(r-1)$ parameters (in fact, one parameter per mixture for $r-1$ distinct variables), we can always avoid the above error. For instance, in the above example, we first set the alive set $A_1 = \{1, 2\}$. But now, instead of using $A_2 = \{3, 4\}$, we will use $A_2 = \{1, 3, 4\}$ (equivalently, $\{2, 3, 4\}$). Then, the recovered tuples under A_2 would be $(0.1, 0.3, 0.4)$ and $(0.6, 0.8, 0.9)$. Since by A_1 we already know the mixture the probabilities 0.1 and 0.6 correspond to, we can recover the other parameters without any errors. Here, r was 2 and we only needed to carry $r(r-1) = 2$ parameters 0.1, 0.6. Formally, we have

Lemma H.2 (Carry overhead is small). *Given $f = \sum_{i=1}^r \mu_i \prod_{j=1}^n (\alpha_{ij} x_j + 1)$, assuming a correct reconstruction algorithm, we can recover the correct distribution by requiring an overhead of at most $r(r-1)$ parameters (equivalently, one parameter value for each mixture for each of the $r-1$ distinct variables).*

Proof. Without loss of generality, suppose there is a subset $S \subseteq [r]$ of size k such that

$$S = \{1, 2, \dots, k\} \quad \text{and} \quad \mu_1 = \mu_2 = \dots = \mu_k = \mu.$$

Fix an alive set A of size $m \geq r+1$. By relabeling variables, we may assume $A = [m]$.

We claim that there exists a subset $V \subseteq A$ of size $|V| = k-1$ such that for every $i \in S$, and for every $\ell \in S \setminus \{i\}$, there exists some $j \in V$ with

$$\alpha_{ij} \neq \alpha_{\ell j}.$$

In other words, by examining the parameters on the coordinates indexed by V , we can distinguish all k mixture components in S .

For $k=1$ there is nothing to prove. For $k=2$, Proposition H.1 implies that the two components must differ on at least one coordinate $j \in A$, i.e., $\alpha_{1j} \neq \alpha_{2j}$. By permuting coordinates, assume $j=1$. Thus, to distinguish the two mixtures it suffices to retain the single variable x_1 , and hence $T_2 \leq 1$.

Let T_k denote the minimum number of variables (coordinates) needed to distinguish k mixture components whose weights are all equal (i.e., $\mu_1 = \dots = \mu_k$). We have $T_1 = 0$ and $T_2 \leq 1$. Assume inductively that

$$T_\ell \leq \ell - 1 \quad \text{for all } \ell < k.$$

We now prove $T_k \leq k-1$.

By Proposition H.1, there exists at least one coordinate $b \in A$ such that not all of $\alpha_{1b}, \dots, \alpha_{kb}$ are equal (otherwise all k components would be identical). By permuting coordinates, assume $b=1$. Consider the multiset $\{\alpha_{i1}\}_{i \in S}$ and partition S into buckets $B_1, \dots, B_g$ such that two indices $i, i' \in S$ lie in the same bucket iff $\alpha_{i1} = \alpha_{i'1}$. Then $g \geq 2$ and $\sum_{t=1}^g |B_t| = k$.

The coordinate x_1 separates different buckets, but does not distinguish components within the same bucket. Therefore, to fully distinguish all k components, it suffices to: (i) keep coordinate x_1 , and (ii) within each bucket B_t , keep $T_{|B_t|}$ additional coordinates to distinguish its members. Hence,

$$T_k \leq 1 + \sum_{t=1}^g T_{|B_t|}.$$

Using the inductive hypothesis $T_{|B_t|} \leq |B_t| - 1$, we obtain

$$T_k \leq 1 + \sum_{t=1}^g (|B_t| - 1) = 1 + \left(\sum_{t=1}^g |B_t| \right) - g = 1 + k - g \leq k - 1,$$

since $g \geq 2$. This proves the claim that $k-1$ variables suffice to distinguish the k components in S .

Finally, in the general case we can partition $[r]$ into $h \geq 1$ buckets $S_1, \dots, S_h$, where within each S_j all mixture weights are equal. Applying the above argument to each bucket, we can distinguish the components inside S_j using at most $|S_j| - 1$ variables. Therefore, the total number of variables we need to carry over is at most

$$\sum_{j=1}^h (|S_j| - 1) = r - h \leq r - 1.$$

Since our initial alive set has size $m \geq r + 1$, we can choose a subset $A' \subseteq A$ of size at most $r - 1$ containing these distinguishing coordinates, and thereafter ensure that every subsequent alive set contains A' . This guarantees that across projections we never lose the ability to distinguish mixture components, and we can repeat the procedure until we recover all parameters α_{ij} for all $j \in [n]$. $\square$

Recall that to properly reconstruct the third-order moments we needed $m \geq 3r + 2$. Now, to ensure we don't confuse between the parameters Lemma H.2 tells us that we need $r - 1$ more variables. Thus, with $m = (3r + 2) + (r - 1) = 4r + 1$, we can always recover the parameters without any issues.